

Name: _____



Level 1 — Positions, Motions, Letters & Words

Created by Austen Cloud · v1.0

Read Me First

You've come across The Kinetic Alphabet, a notation system designed to help you craft and communicate your own unique choreography. This grid-based language is designed for music, using pictographs and letters that combine like puzzle pieces for each beat. This system has propelled my sequence creation to new heights, and I hope it will do the same for you!

The Kinetic Alphabet is a fusion of elements from VTG (Vulcan Tech Gospel), siteswap (Juggling Notation), and musical notation. Although it can be introduced to beginners, it's designed for intermediate learners, bridging the gap between improvisation and choreography. Originally built for double staves, it can be applied to any dual wielded static prop like clubs, fans, triads, buugeng, and more.

Pictographs form the core of The Kinetic Alphabet. The letters are a useful tool to categorize and communicate the pictographs, but they are secondary to the

pictographs themselves. It's not necessary to memorize the letters immediately to benefit from this system.

In a pictograph, the arrow shows the direction of a hand's motion and the hand marker shows where that hand ends. Read each grid as a single beat.

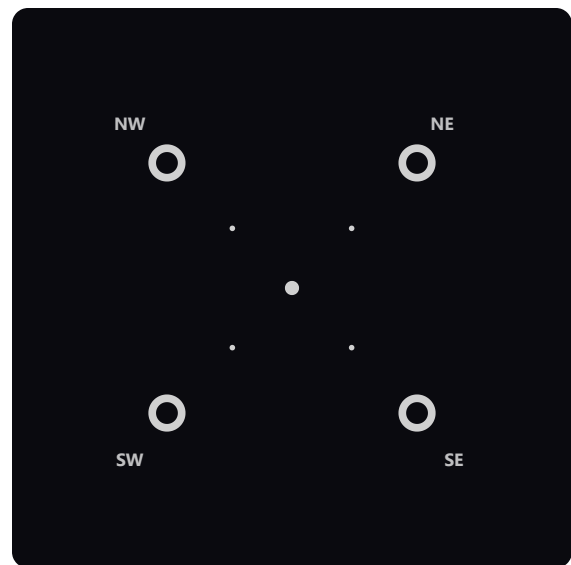
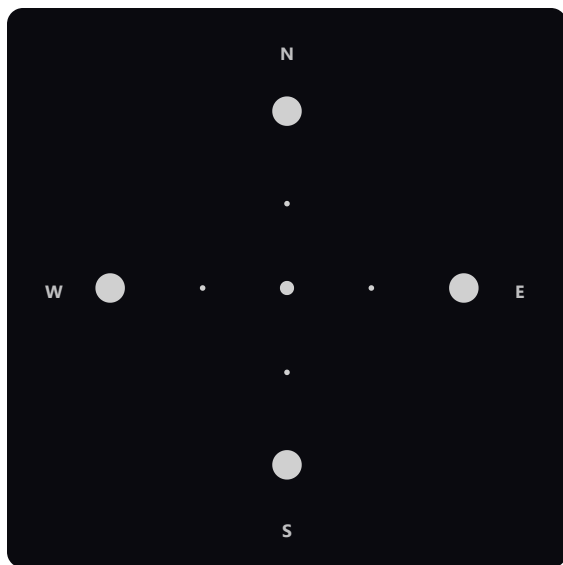
*With love,
Austen Cloud*

Positions & Motions

The Grid

The Kinetic Alphabet is based on a 4-point grid.

There are two 4-point grids: box mode and diamond mode. This guide is written in diamond, but everything translates to box.



On this grid, there are three types of points:

The center point is the hub that everything revolves around.

The four hand points are halfway between the center point and the outer points.

The outer points depict the outer edges of the grid.

Together, diamond and box form an 8-point grid. We'll use diamond mode to learn each concept.



Hand Positions

There are multiple ways to combine two hand points to form a hand position. Positions can be rotated or mirrored. Red = Right and Blue = Left.

In The Kinetic Alphabet, our first three positions are called Alpha, Beta, and Gamma.

Alpha

In Alpha, the hands occupy the points across from each other.

Alpha
Hands across from each other



Beta

In Beta, the hands occupy the same point.

Beta
Hands at the same point



Gamma

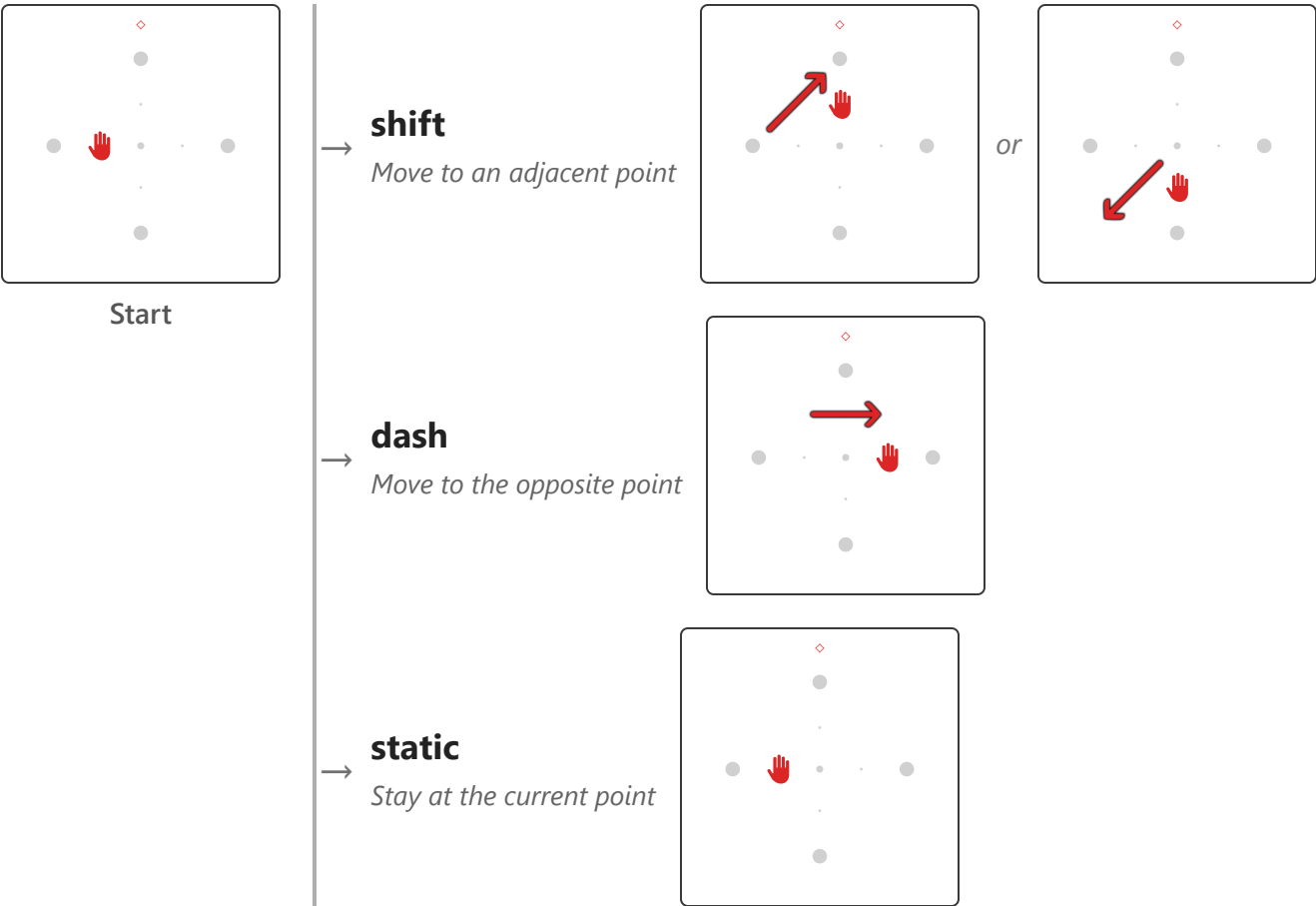
In Gamma, the hands form a right angle.

Gamma
Hands form a right angle



Hand Motions

There are three fundamental hand motions in the Alphabet. The arrow shows the direction of motion. The hand shows the end position.



Using these, we can derive six combinations, named below. We'll explore each one individually:

Dual-Shift Both hands travel to an adjacent point.	Shift One hand travels to an adjacent point and the other hand remains static.
Cross-Shift One hand travels to an adjacent point and the other travels to the opposite point.	Dash One hand travels to the opposite point and the other hand remains static.

Dual-Dash

Both hands travel to the opposite point.

Static

Both hands remain static.

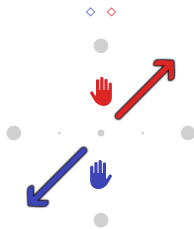
Type 1 - Dual-Shifts

When both hands shift to adjacent points at the same time, the combination is called a **Dual-Shift**.

Split-Same (Alpha to Alpha)

When both hands start apart (alpha) and end apart (alpha), they travel in parallel paths between two pairs of adjacent points.

Both hands shift (same direction)

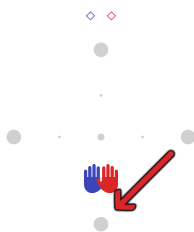


Dual-shift: both hands shift in parallel, alpha to alpha

Together-Same (Beta to Beta)

When both hands start together (beta) and end together (beta), they shift from one shared point to another shared point.

Both hands shift from east to south

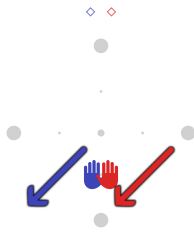


Dual-shift: both hands shift together from east to south, beta to beta

Alpha and Beta (Split to Together)

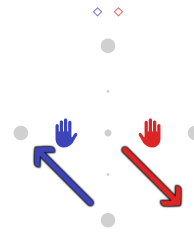
Hands can also shift between alpha and beta. Starting apart, both shift toward the same point and end together, or vice versa.

Split to together



Dual-shift: hands start apart and end together

Together to split



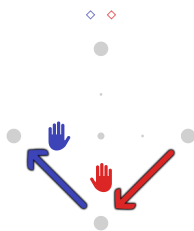
Dual-shift: hands start together and end apart

Gamma

When both hands start at a right angle to each other (gamma), dual-shifts create a different set of movements.

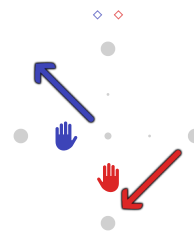
In gamma, one hand is at a cardinal point and the other is at an adjacent cardinal point, forming a 90-degree angle through the center.

Gamma to gamma



Dual-shift from gamma to gamma

Gamma (opposite direction)

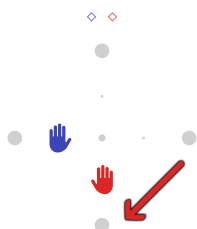


Dual-shift at gamma, hands moving in opposite directions

Type 2 - Shifts

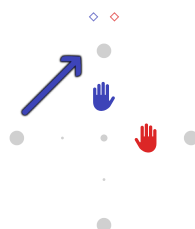
A **Shift** (capital S) is when one hand shifts to an adjacent point while the other hand stays still. Shifts move you between gamma and alpha/beta.

Red shifts, blue stays



Shift: right hand shifts while left hand stays static

Blue shifts, red stays

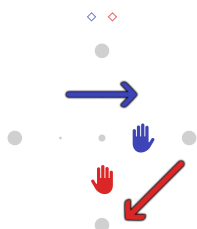


Shift: left hand shifts while right hand stays static

Type 3 - Cross-Shifts

A **Cross-Shift** combines a shift and a dash. One hand shifts to an adjacent point while the other dashes straight through the center to the opposite point.

Red shifts, blue dashes

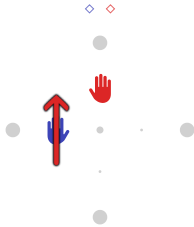


Cross-shift: right hand shifts while left hand dashes across

Type 4 - Dash

A **Dash** (capital D) is when one hand dashes to the opposite point while the other stays still.

Red dashes, blue stays

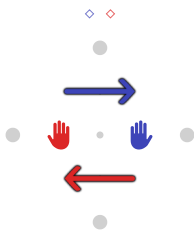


Dash: right hand dashes across while left hand stays static

Type 5 - Dual-Dash

A **Dual-Dash** is when both hands dash simultaneously to their opposite points.

Both hands dash through center



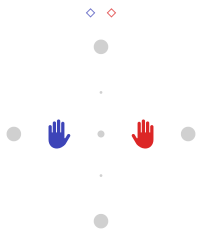
Dual-dash: both hands dash across the center

Type 6 - Static

Static means both hands remain at their current positions. No motion occurs.

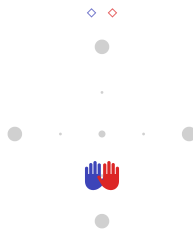
Later, static positions gain complexity when adding prop rotations.

Static at alpha



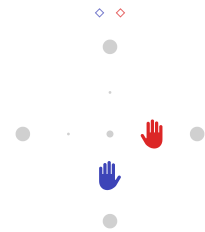
Static: both hands hold at alpha
(opposite points)

Static at beta



Static: both hands hold at beta (same
point)

Static at gamma



Static: both hands hold at gamma
(right angle)

Staff Positions

When writing sequences with staves, it helps to mark the thumb end with a line. The performer can use it to keep track of rotations and check their position on every beat. It also encourages negative space/body turns instead of finger spinning.

In the following examples, an end is always at the center point. Practice each position below, paying attention to the thumb orientation.

Many pictographs in this guide are depicted with no thumb ends when categorizing. It is usually noted only during sequences.

Most sequences in this guide start with thumbs in for consistency. It's equally valid to start any sequence from a different thumb orientation.



Staff positions showing thumb in and thumb out orientations at Alpha, Beta, and Gamma

Staff positions - thumbs in and out at Alpha, Beta, and Gamma

Staff Motions

During a shift, a prop can rotate in one of two directions - Prospin or Antispin.

Prospin

The prop rotates the same direction as the handpath. A 90 degree isolation is our base unit of a prospin.

In a base isolation, the thumb orientation remains the same for the entire motion.

Antispin

The prop rotates in the opposite direction of the handpath. A 90 degree antispin is our base unit of antispin.

In an antispin, the ends swap orientation. It moves from thumb in to thumb out.

Halfway through the motion, the center of the staff is at the grid's center point.

In a base dash, the thumb ends also swap orientation.



Prospin: prop and hand rotate same direction. Antispin: opposite directions.

Prospin vs antispin

Negative Space / Body Turns

Many sequences seem impossible, but most can be solved by using negative space or body turns.

Negative space lets you face the audience and reduces body movement.

Body turns add movement and help you execute patterns with longer staves.

Each method is equally important, and learning both will maximize capability.

This guide will assume some knowledge of these fundamental concepts.

To make the most of the Alphabet, it's highly recommended that you learn the following.

4-Petal Antispin

To execute this in wall plane, you must do one of the following on beat 2:

Pass the thumb end through the negative space above your right shoulder on beat 2.

Turn your torso to the left on beat 2 and pass the thumb end in front, then pass the pinky end on the inside of your right arm as you move to beat 3.

Practice in reverse, then do both directions in the other hand. Then practice everything again starting with the thumb out. Try using both negative space and turns. Good luck!

360° Isolation

To execute this without finger-spinning, turn your torso to the left on beat 3. During this beat, the staff moves briefly in wheel-plane relative to your left-facing view. On beat 4, turn your body back to center as you return to the start position.

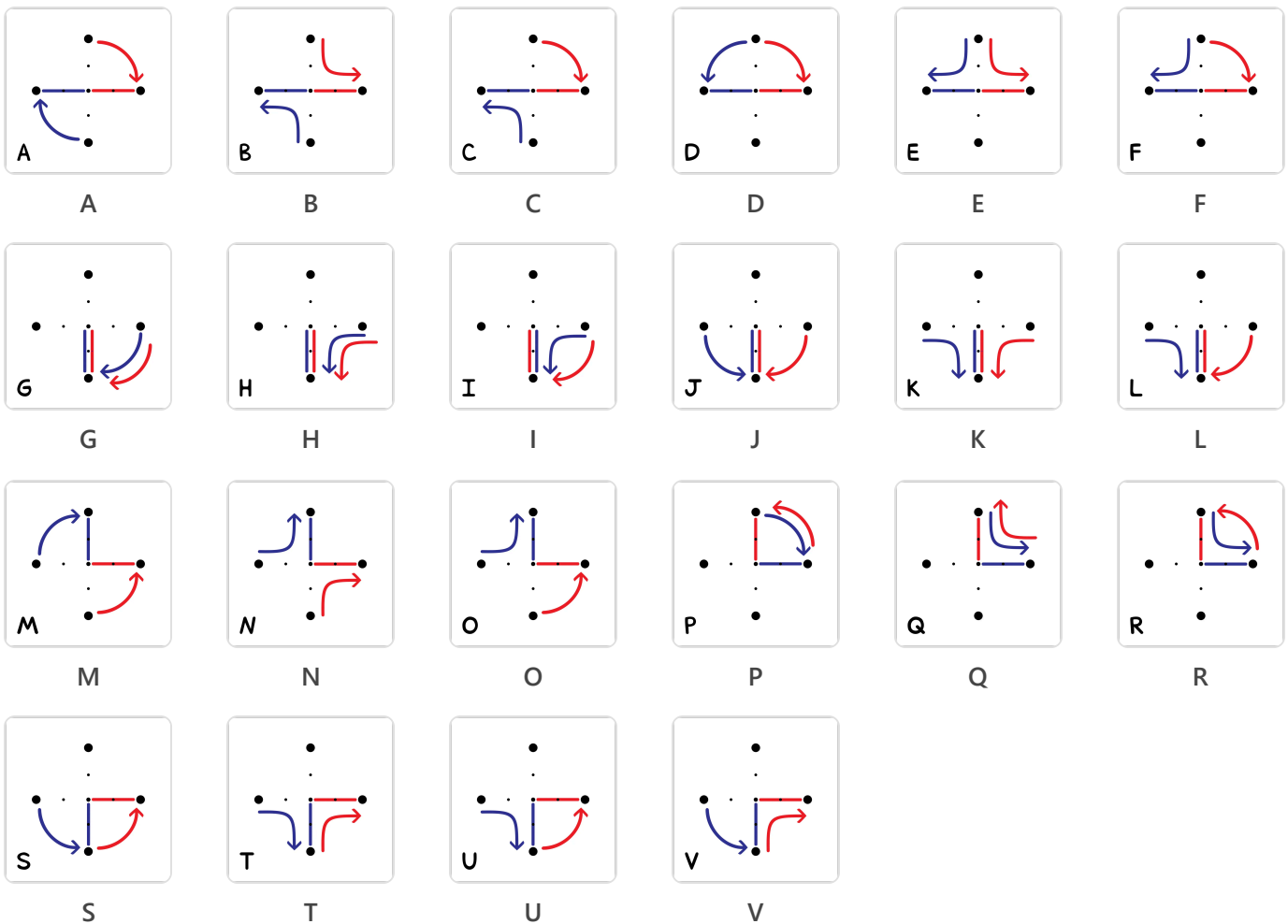
Practice in reverse, then do both directions in the other hand. Then practice it with the thumb out, isolating the pinky end.

Letters

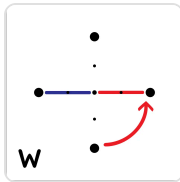
Double Staff Codex

Just like positions, each motion pictograph can be rotated, reflected, or color swapped. Letters are organized on the page by end position, Alpha, Beta, then Gamma. Let's look at each type individually.

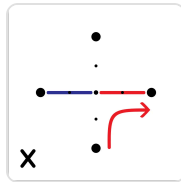
Type 1 — Dual-Shift



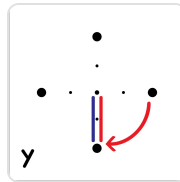
Type 2 — Shift



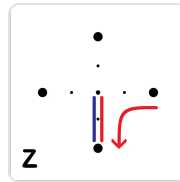
W



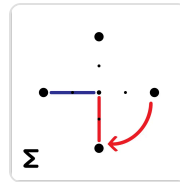
X



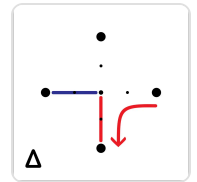
Y



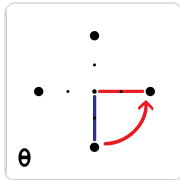
Z



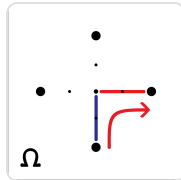
Σ



Δ



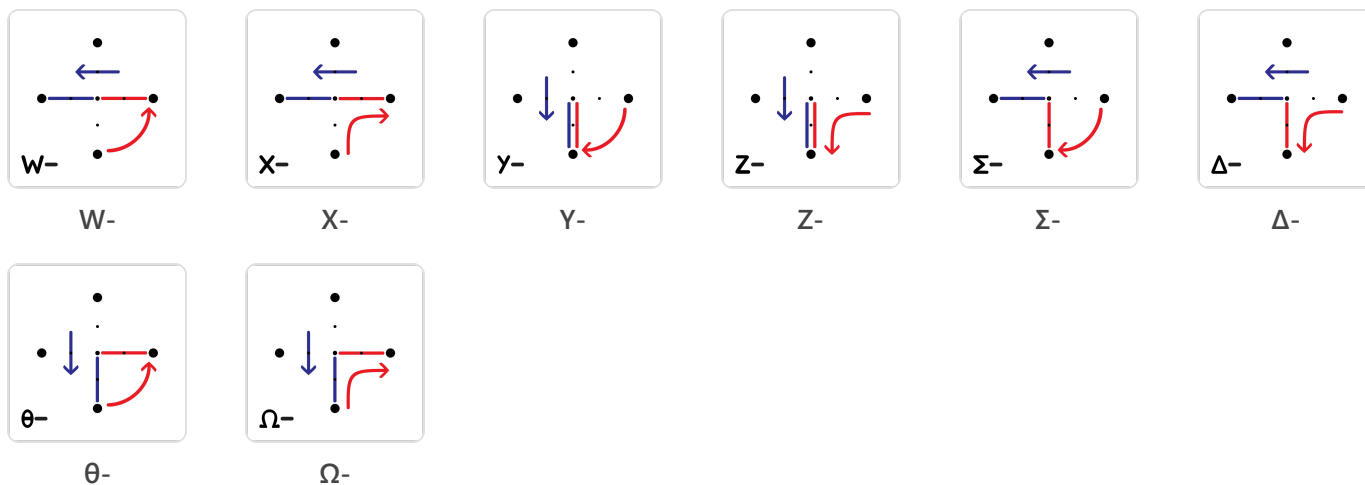
θ



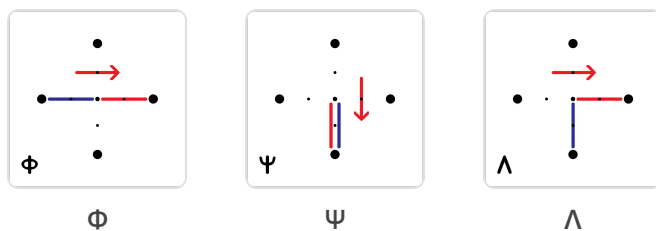
Ω

Codex — Types 3–6

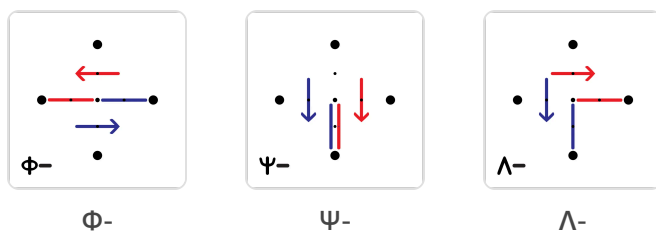
Type 3 — Cross-Shift



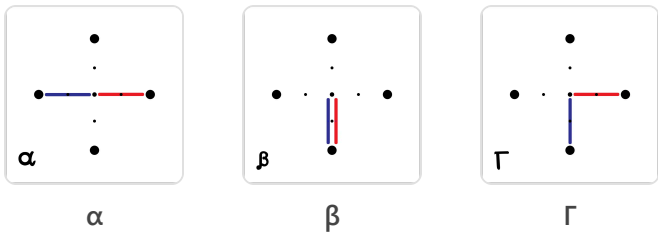
Type 4 — Dash



Type 5 — Dual-Dash

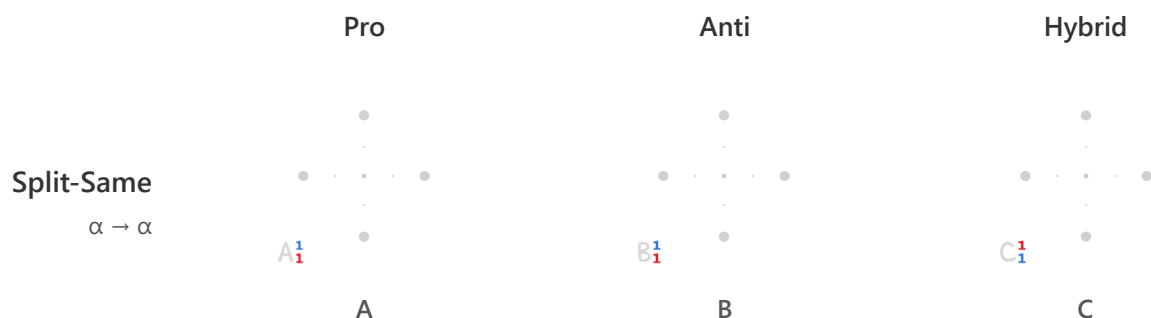


Type 6 — Static



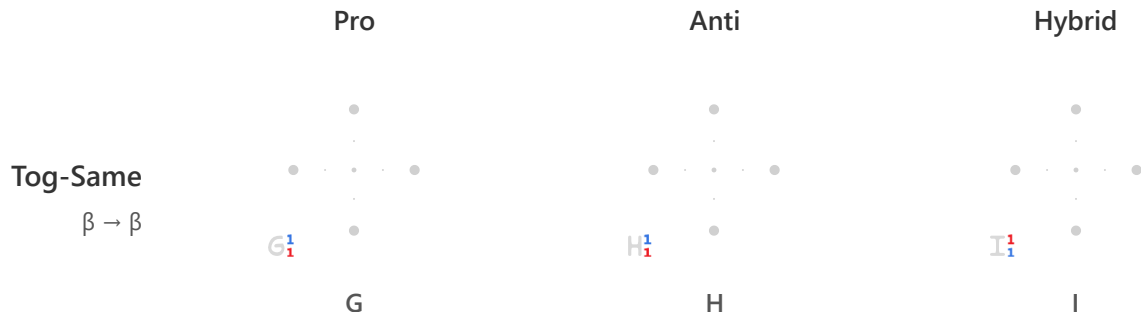
Type 1 — Dual-Shift Letters

First we'll look at A, B, and C. Their handpath is Split-Same and they move from $\alpha \leftrightarrow \alpha$:



Notice the pattern: Pro - Anti - Hybrid. This pattern helps you navigate/memorize the letters. If you only remember that A has prospins, you can infer that B has antispins. If you only remember that B has antispins, you can infer that C is a hybrid. If you memorize only one letter in each group, you know all of them.

Next let's look at G, H, and I. Their handpaths are Tog-Same and they move from $\beta \leftrightarrow \beta$:



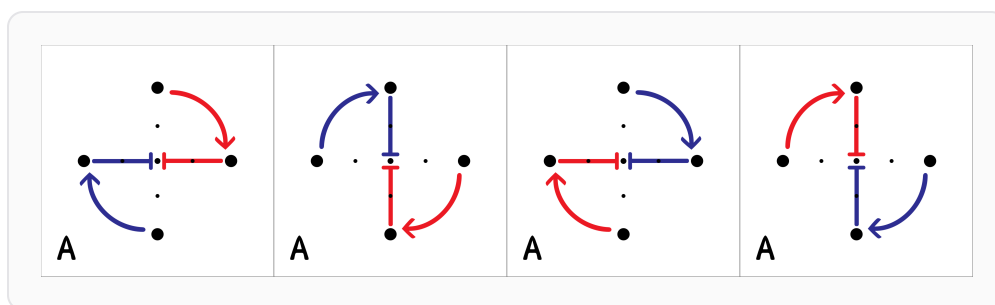
In hybrids like C and I, either hand can execute a prospin or antispin. Here, the right is in pro and left in anti, but it's equally valid to swap this.

Alpha/Beta Words

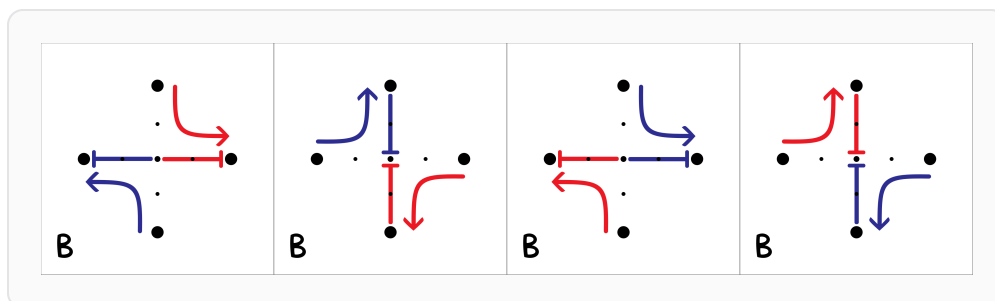
Same Direction

The first words we will learn correspond to VTG's 1:1 motions. To execute these, you'll need to use body turns and/or negative space. Practice each word once in both directions, then again starting with thumbs out.

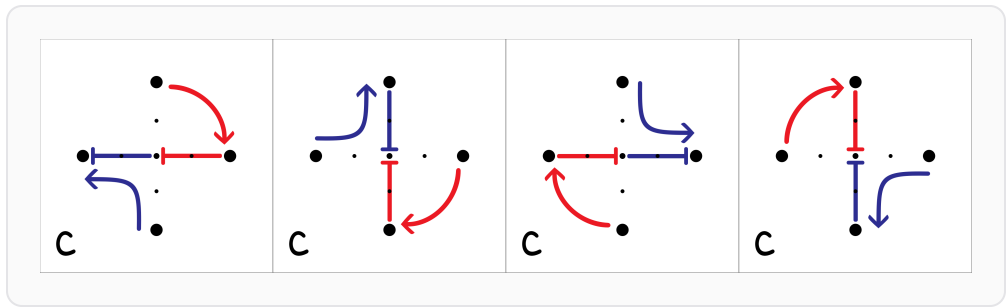
Split-Same



AAAA

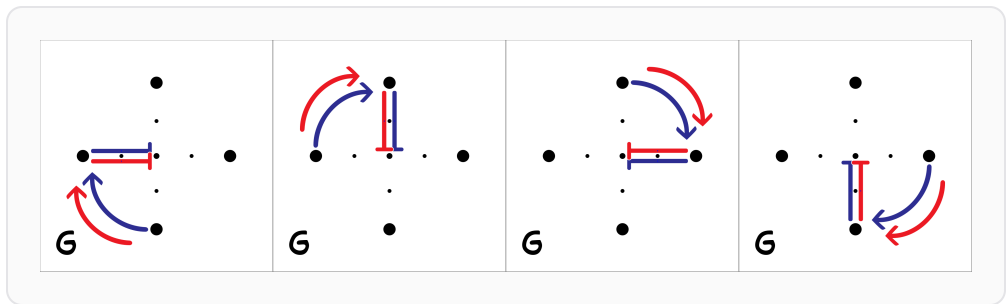


BBBB

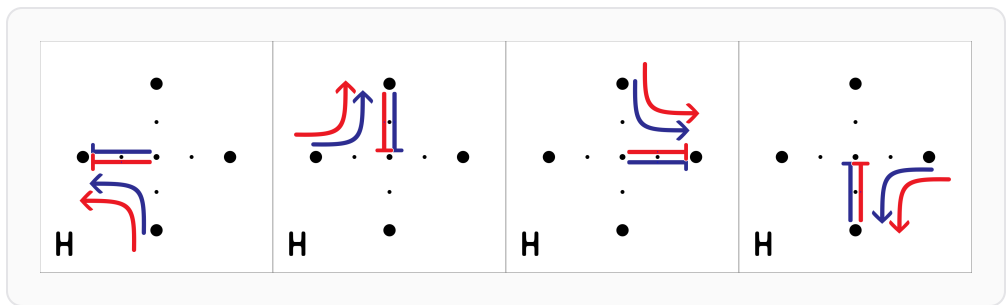


CCCC

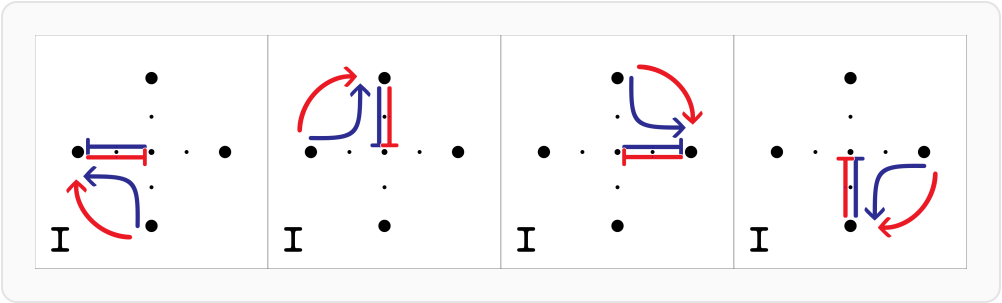
Tog-Same



GGGG



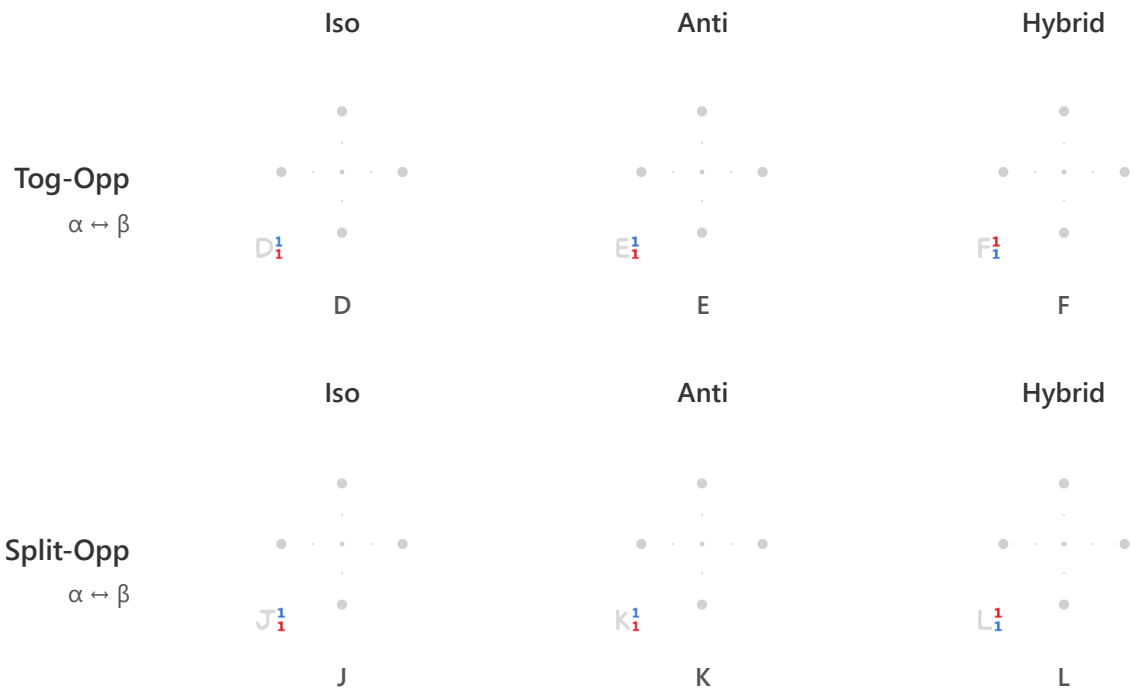
HHHH



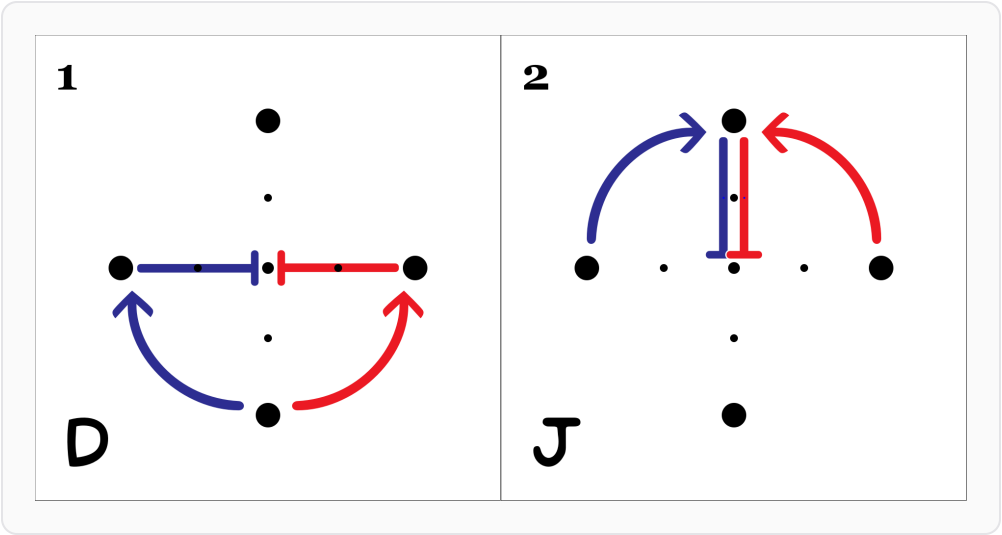
||||

Compound Letters

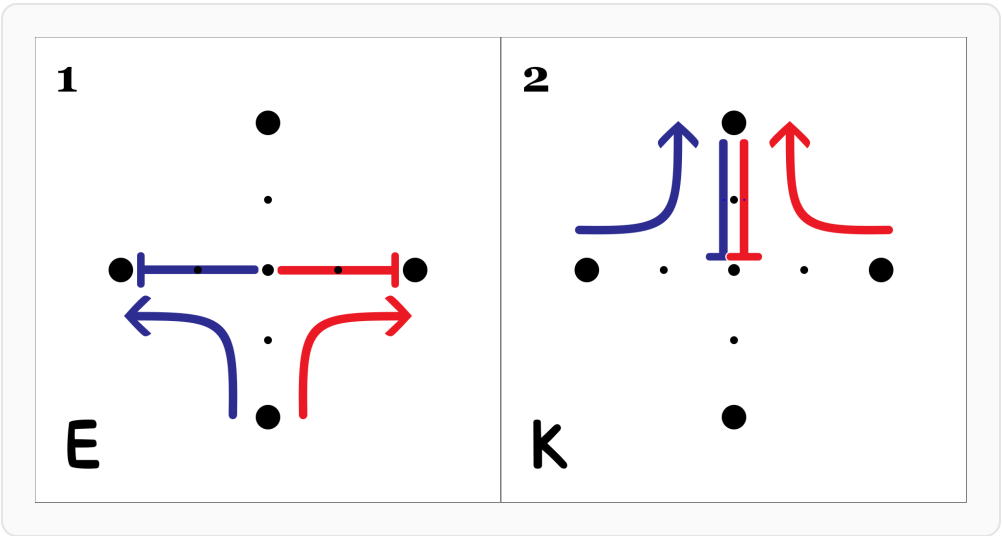
Now let's look at the letters that move from $\alpha \leftrightarrow \beta$ or $\beta \leftrightarrow \alpha$. All pictographs can be rotated or mirrored without changing letters. These can be either Tog-Opp or Split-Opp depending on which α/β you start from.



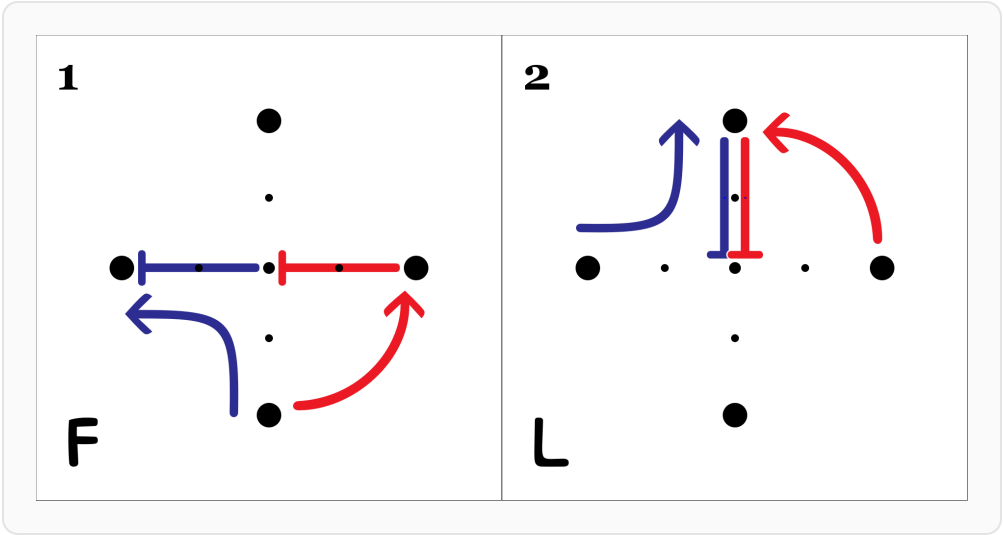
These compound letters can't be self-combined like the previous letters. Instead, they combine with other compound letters to form the words DJ, EK, and FL. Here they are along with cute phrases to help you remember:



DJ — Disco Jam



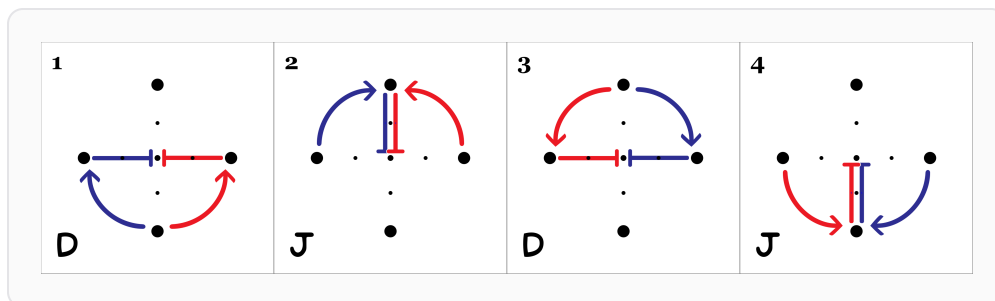
EK — Exploding Kitten



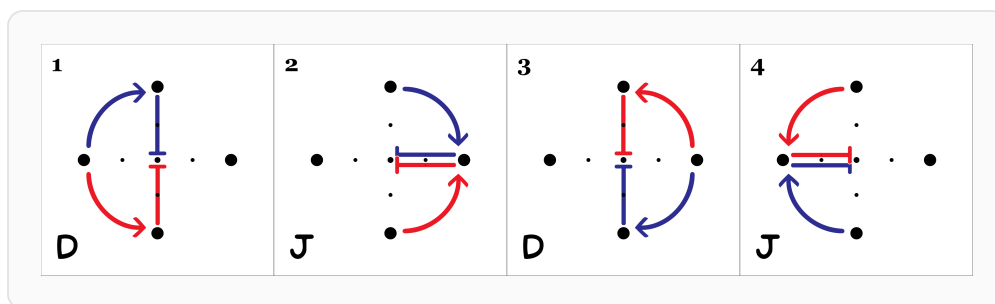
FL — Fruity Loops

Compound Words

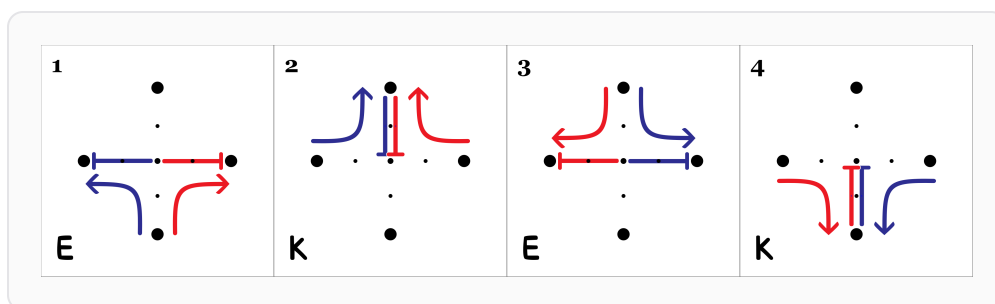
These are the simplest words that use the compound letters DJ, EK, and FL. For Tog-Opp, cross your arms or body turn to use the plane behind you. For Split-Opp, either body turn or use negative space. Both versions are equally valid. Try them yourself!



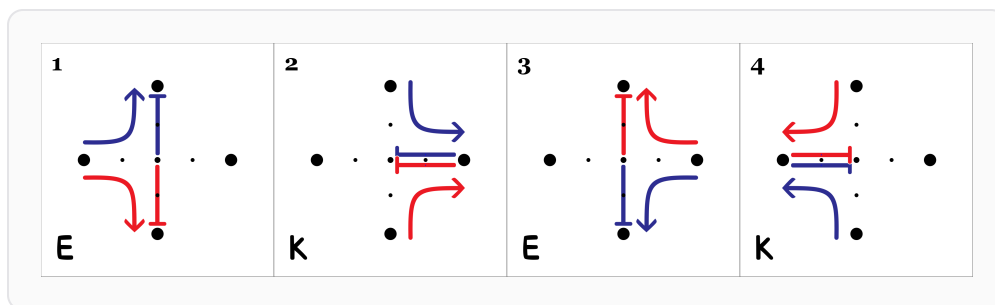
DJ — Tog-Opp



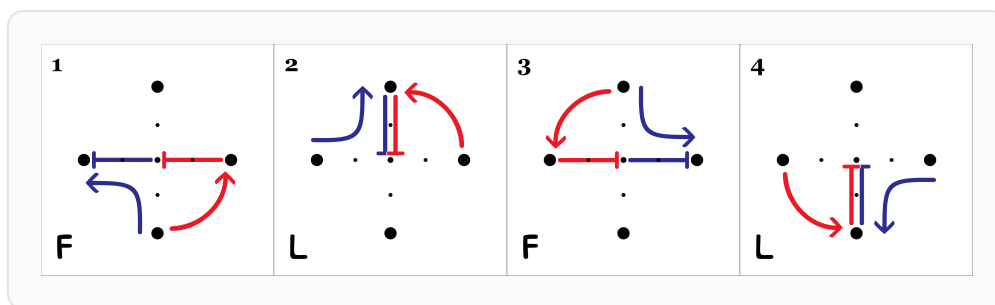
DJ — Split-Opp



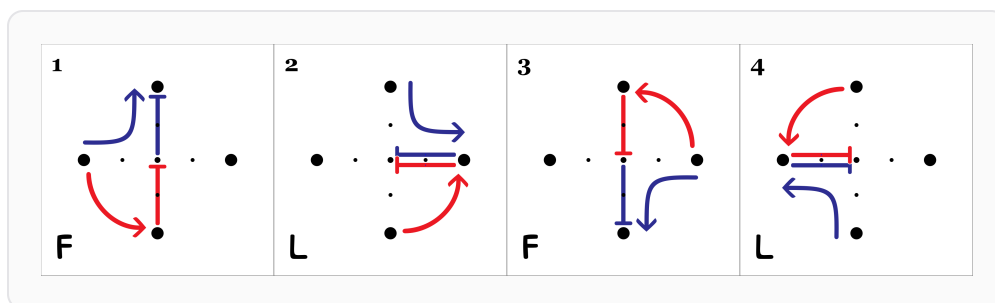
EK — Tog-Opp



EK — Split-Opp



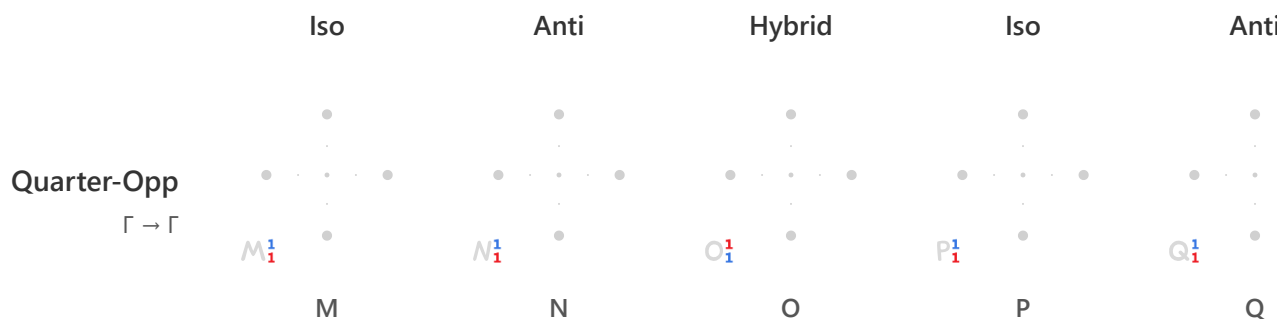
FL — Tog-Opp



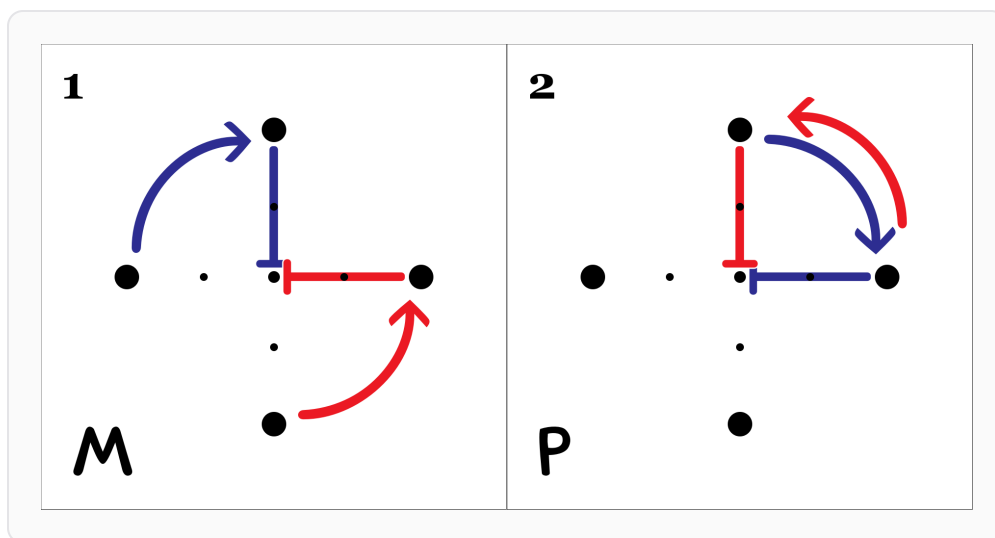
FL — Split-Opp

Gamma Letters

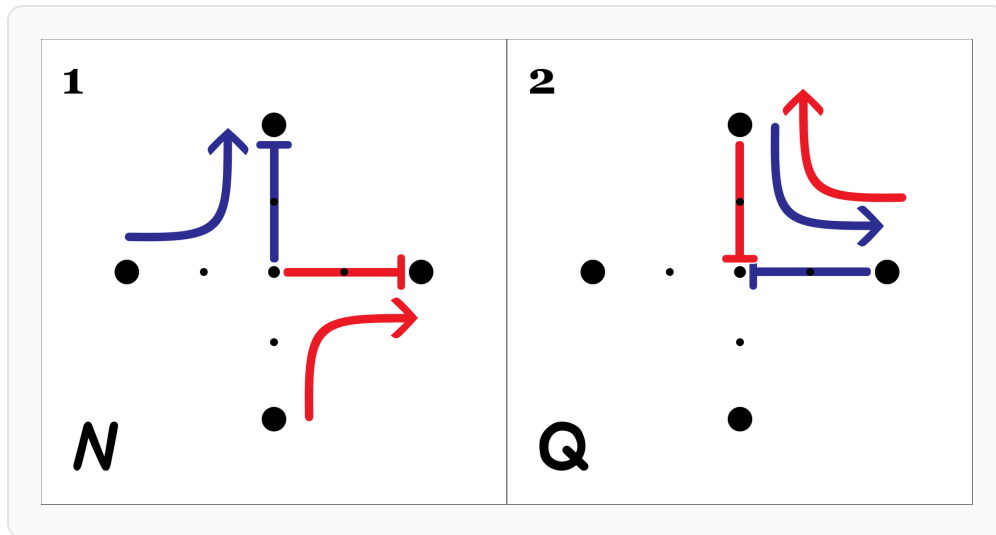
Γ motions can combine with any other Γ motion to create lots of words! First let's look at the compound letters (Quarter-Opp).



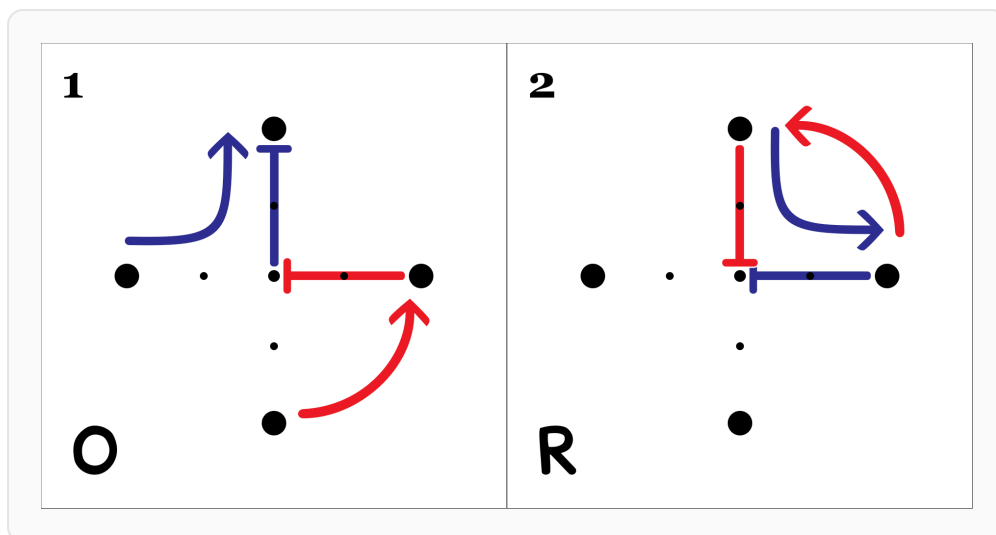
When combined as a continuous motion, these form MP, NQ, and OR. Here they are along with a memorable phrase:



MP — Magic Potion



NQ — Never Quit



OR — Open Road

The final Γ group (Quarter-Same) has 4 instead of 3. It may seem like U and V contain the same information, but it's impossible to rotate or reflect U in order to turn it into V, and vice-versa, so they must be disambiguated.

S

T

U

V

Quarter-Same

$\Gamma \rightarrow \Gamma$

S¹₁

S

T¹₁

T

U¹₁

U

V¹₁

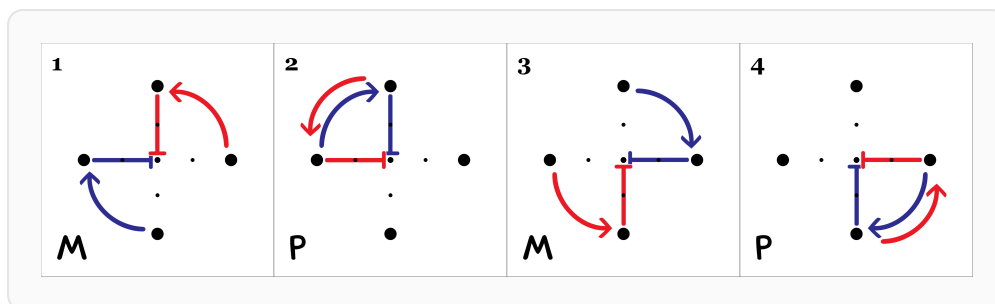
V

Note that all four have a leading hand and a following hand. Here, the right is leading and left is following, but it's equally valid to swap this. U leads with an isolation (a round motion like the letter U). V leads with an antispin (a spiky motion like the letter V). These self-combine to form the words SS, TT, UU, and VV.

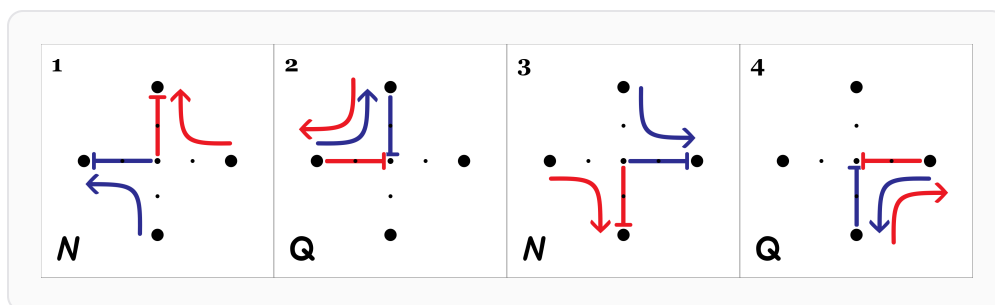
Gamma Words

These are the simplest 4-letter words created with continuous Γ motions:

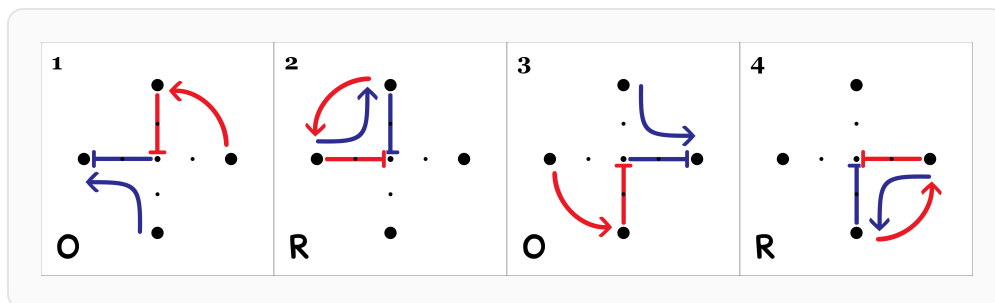
Opp



MP

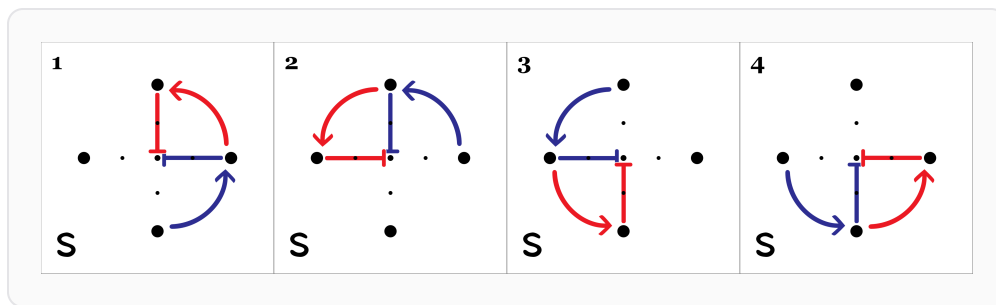


NQ

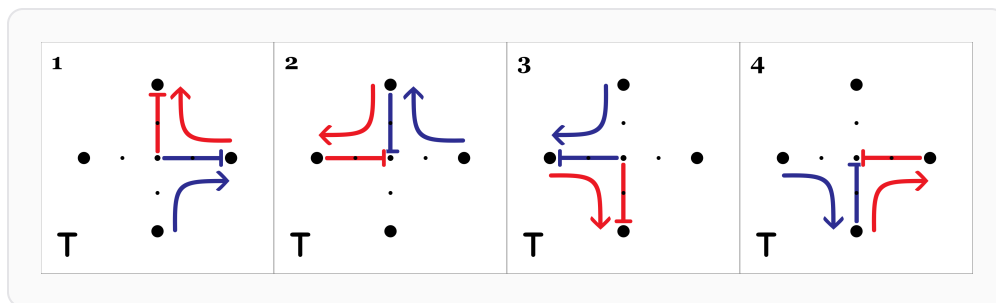


OR

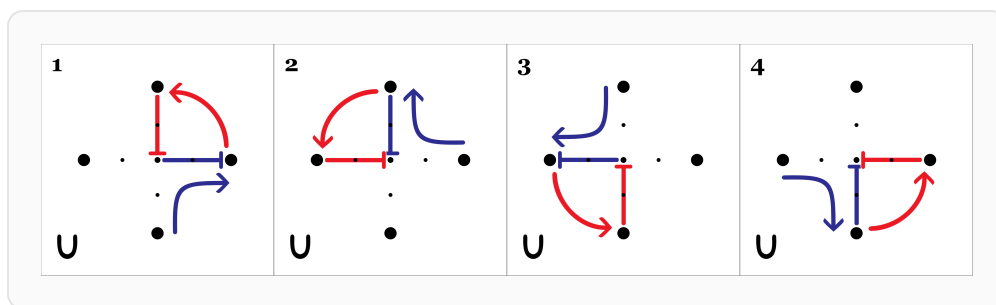
Same



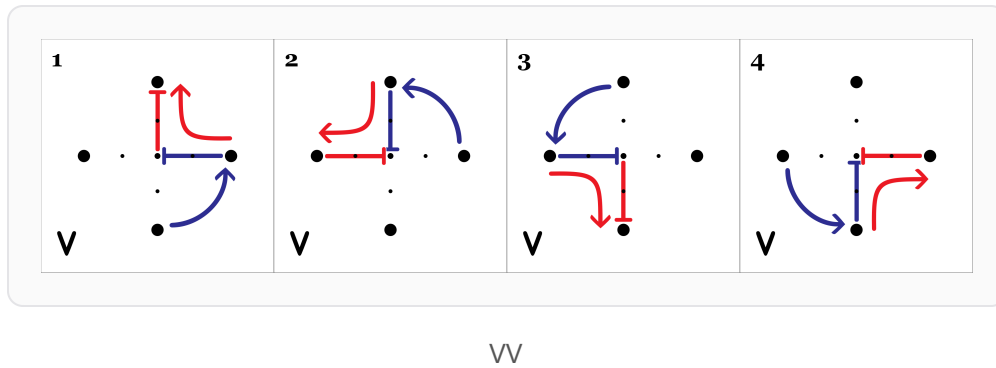
SS



TT



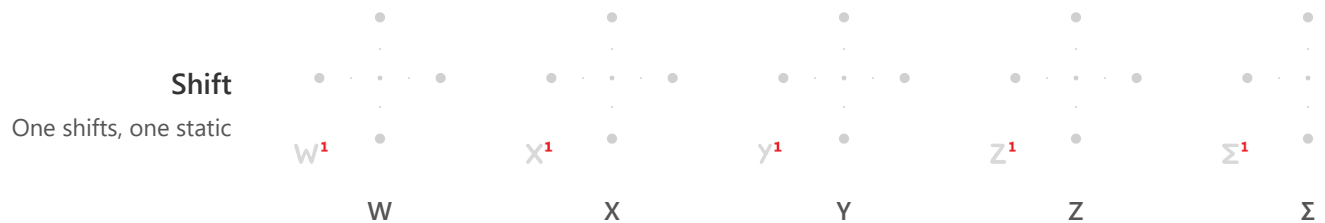
UU



Type 2 — Shift Letters

So far we've learned how to move between $\alpha \leftrightarrow \alpha$ and between $\beta \leftrightarrow \beta$ and between $\alpha \leftrightarrow \beta$. In order to travel between these two modes, we can use a Type 2 Motion called a Shift.

A Shift (or single shift) is the combination of one shift and one static motion. Their letters are organized by end position: α , β , then Γ . These can also be categorized by opening or closing.

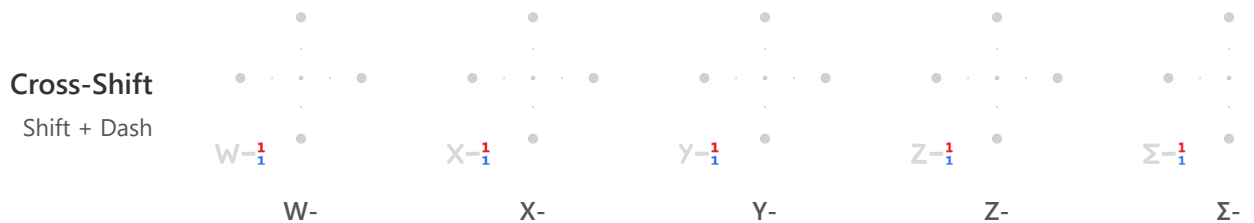


When we arrange them in continuous motions, we get the words WΣYΘ and XΔZΩ.

Though simple at this stage, these motions become more complex as we dive deeper into the Alphabet and add rotations to static motions.

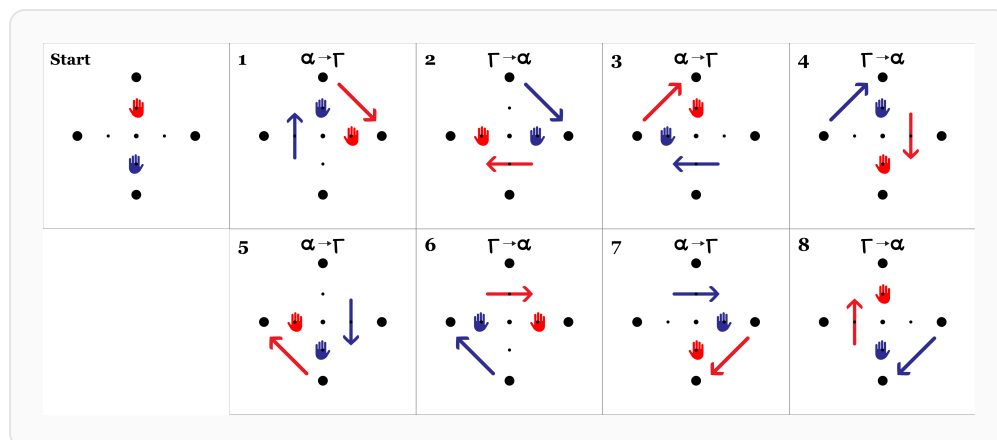
Type 3 — Cross-Shift Letters

Cross-Shifts use the same letters as Shifts, but each letter is followed by a dash to indicate that the other hand is dashing into its end position. They are spoken as "W Dash" or "Sigma Dash". A dash symbol in the glyph equals a dash arrow on the graph. The end position for each Type 2/3 letter remains the same.

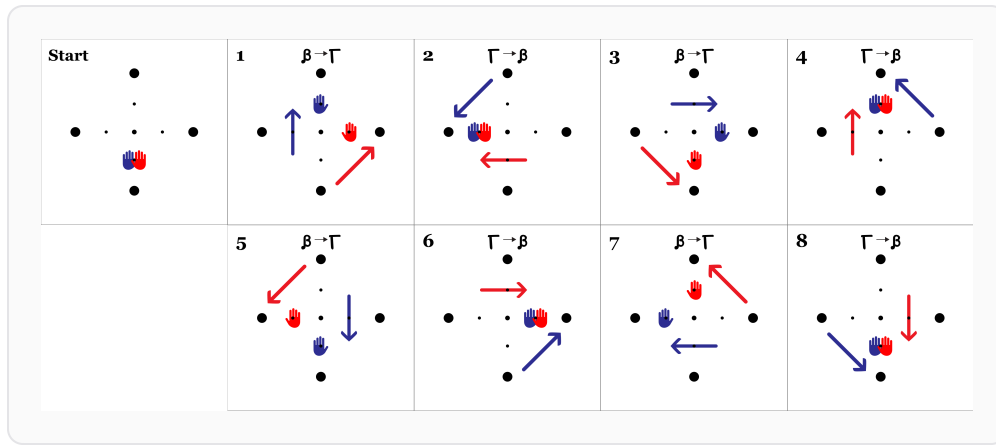


Cross-Shifts can be tricky to remember. It helps to first picture the corresponding Type 2 pictograph, then add the dash arrow without changing any other variables.

Just like we did with hands, let's break down some Cross-Shifts step-by-step:



Alpha Cross-Shifts — start → halfway → end



Beta Cross-Shifts — start → halfway → end

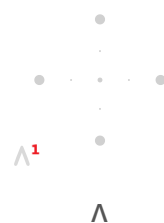
When initially learning, it's useful to pause at the halfway point to ensure proper timing.

Type 4/5/6 Letters

Type 4 — Dash

With a Dash, one prop executes a dash and the other remains static. "Lambda" can be further shortened by calling it "Lam".

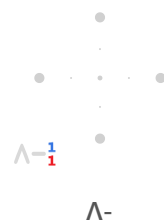
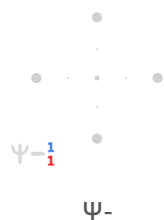
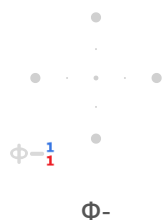
Dash
One dashes, one static



Type 5 — Dual-Dash

In a Dual-Dash, both hands are dashing. The end position remains the same.

Dual-Dash
Both hands dash



Type 6 — Static

In a Static motion, both hands remain still for a beat. These become more interesting when adding turns.

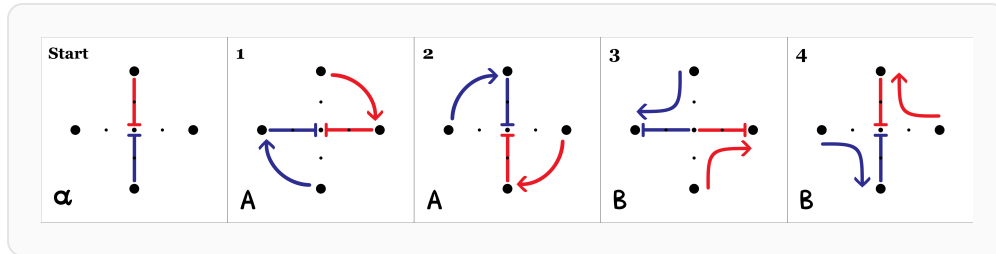
Static
Both hands still



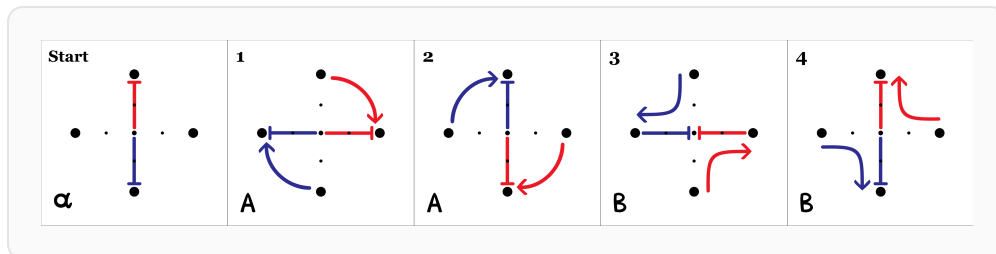
Words, LOOPs & Reversals

Words

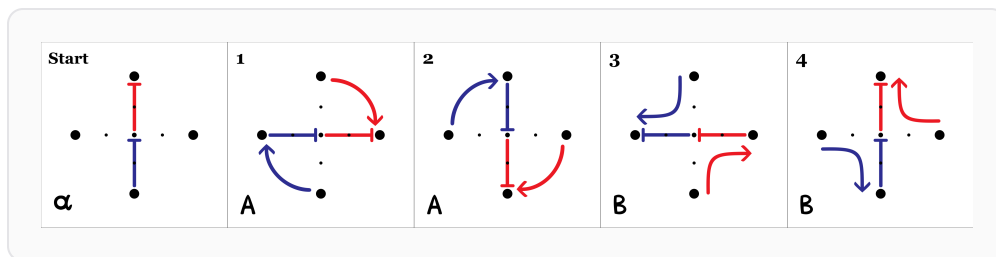
Let's create more complex words using pictographs! In order to perform the words in this section correctly without finger-spinning, you must be familiar with negative space and body turns. If you finger-spin instead of using negative space, you'll lose precision and the ability to check your thumb orientation on each beat to see if you're still on track. We'll use the word AABB as an example. Here are three variations on AABB, starting from different thumb orientations. Use staves or red/blue pens to follow along.



Thumbs: in | in



Thumbs: out | out



Thumbs: in | out

As you execute these with staves, notice that each of these sequences requires a different type of negative space, either above/below the shoulder or behind the elbow. The execution of the same word can feel completely different depending on factors like the start position, rotation direction, and thumb orientation. That's why it's necessary to draw the full sequence with pictographs for complete clarity. The Alphabet is primarily a system of pictographs, organized by letters for convenient communication. The letters do not give all of the information, and are merely intended to separate motion combinations into categories which can be further clarified with detailed pictographs.

LOOPS

When a word ends on a variation of its start position, we can repeat it with a transformation applied, eventually returning back to the start position (aka home). This type of sequence is called a LOOP.

LOOPS have six base transformation types:

Rotated — each repetition's pictographs are rotated (e.g. 90° or 180°) from the previous.

Mirrored — each repetition reflects the previous, changing rotation direction.

Swapped — each repetition swaps the roles of right and left hands.

Flipped — each repetition flips the previous across an axis.

Inverted — each repetition inverts the motion type (pro ↔ anti).

Rewound — each repetition retraces the previous in reverse order.

These base types can combine into compound LOOPS like mirrored-swapped or rotated-inverted.

Rotated

In a rotated LOOP, each repetition ends in a rotated variation on its previous position. In this example, there is a 90° rotation, finally returning to the start position (aka "home").

Mirrored

In a mirrored LOOP, the second repetition's pictographs reflect the first, which changes their rotation direction. In this example, each column is reflected across a horizontal plane.

Swapped

In a swapped LOOP, each repetition swaps the roles of right/left. Though the prop's shapes look the same, this swap changes the body motion significantly.

LOOPS vs CAPs

If you come from the poi community, you may know the term CAP (Continuous Assembly Pattern), coined by Damien (Zaltymbunk) on the Home of Poi forums and popularized by Alien Jon, Nick Woolsey, DrexFactor, and Charlie Cushing. CAPs and LOOPS are parallel concepts that describe cyclic patterns, but they approach composition differently.

How they differ

CAPs compose per-hand trajectories. You overlay a left-hand path with a right-hand path to produce the combined pattern. The most well-known CAP (the "C-CAP") is the kidney-bean shape created by alternating extensions and antispin petals.

LOOPS compose per-beat snapshots. Each letter captures both hands simultaneously for one beat. A LOOP is a word that, when repeated with one or more transformations, returns to its starting position.

Neither is a subset of the other

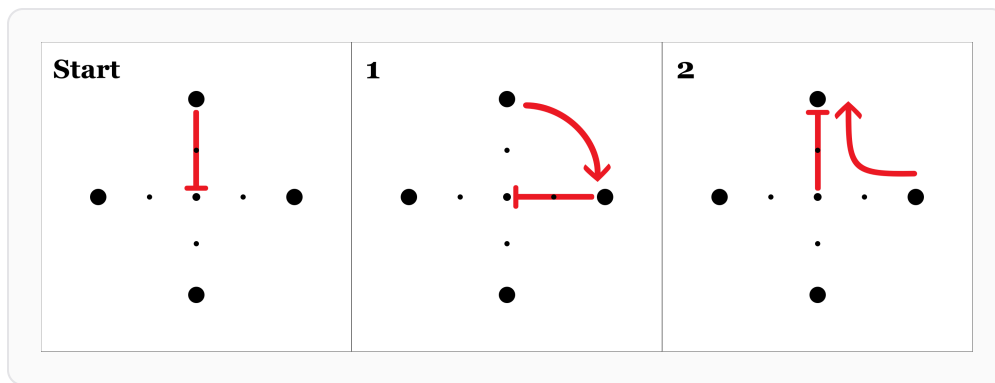
Some patterns can be described as both a CAP and a LOOP. Others exist in one framework but not the other. CAPs describe trajectory shapes; LOOPS describe beat-by-beat letter sequences with transformation rules. Different lenses on the same physical motion.

Reversals

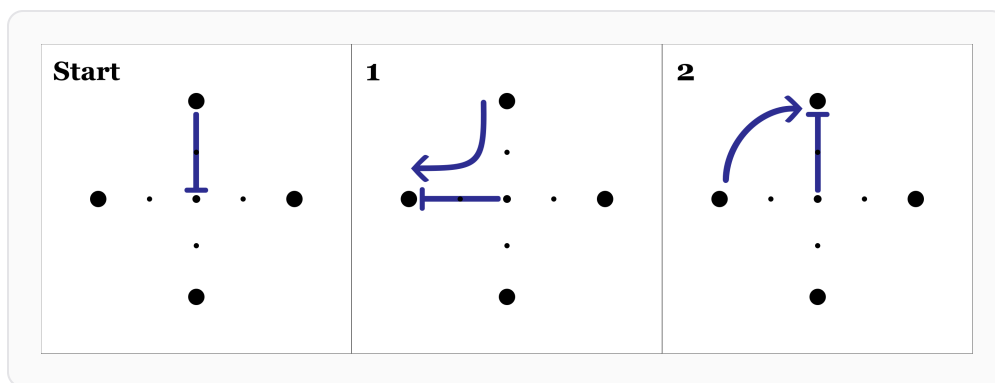
Reversals open up a huge number of possibilities! There are three types of reversals:

Hand-reversal

With a hand reversal, the hand returns to the point it came from previously, without changing the prop's direction of spin. Relative to the center point, this changes a prospin to an antispin and vice-versa. This is the simplest and least disruptive reversal. We've already used it in the previous examples.



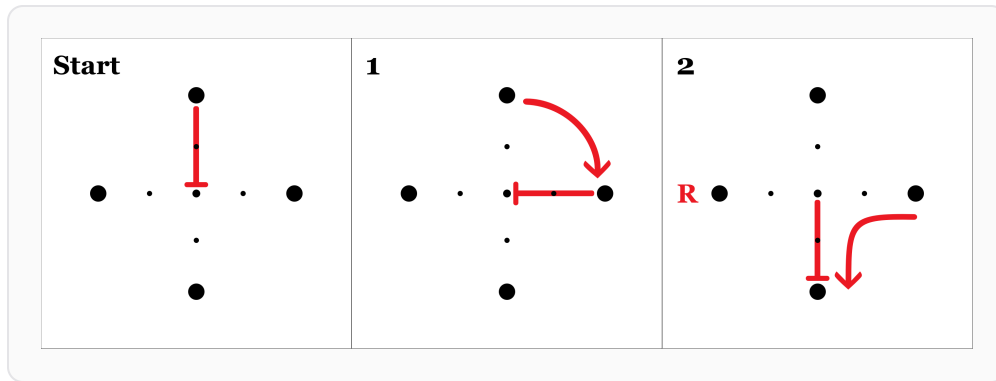
Hand-reversal: pro to anti



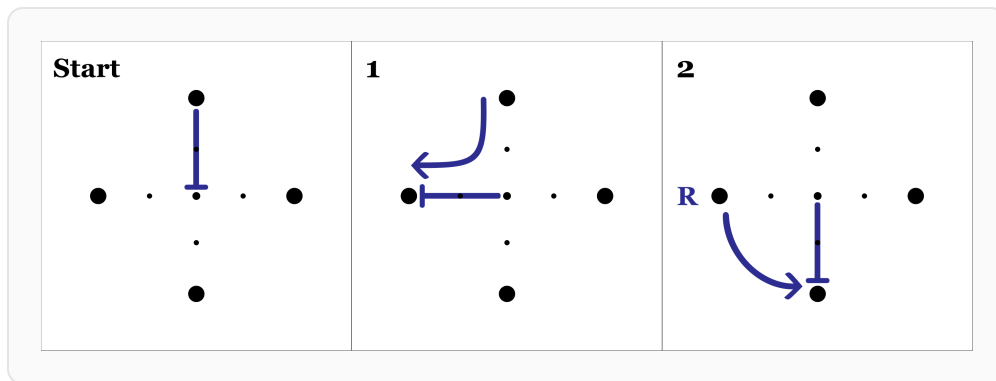
Hand-reversal: anti to pro

Prop-reversal

With a prop reversal, the hand continues to the next point while the prop reverses direction. This reversal also changes a prospin into an antispin and vice-versa. Since a prop reversal is less intuitive, an "R/R" is added in the corresponding color in between the pictographs to indicate it.



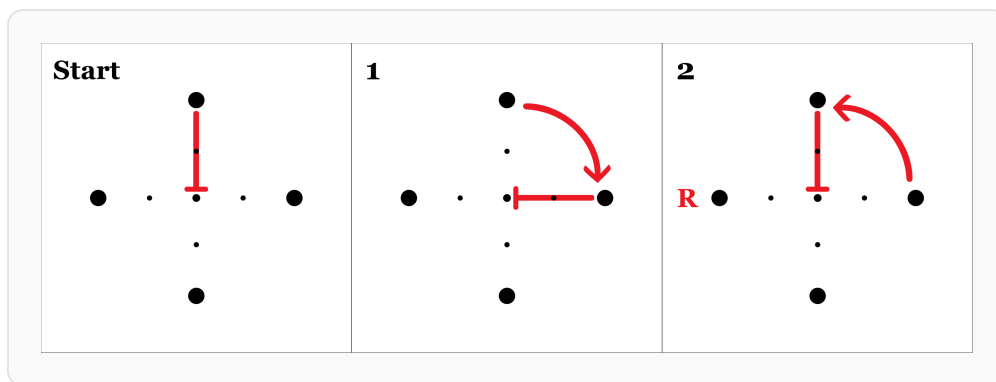
Prop-reversal: pro to anti



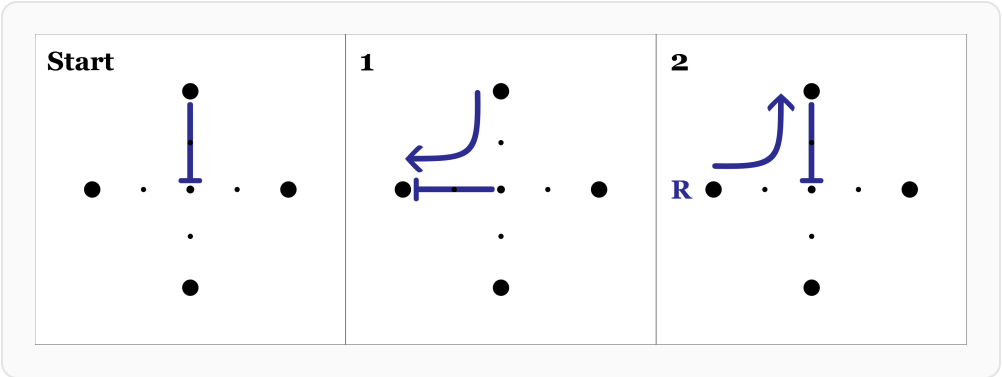
Prop-reversal: anti to pro

Full-reversal

With a full-reversal, the prop and hand retrace their paths and return to their previous position, as if going backwards in time. Because this contains a prop reversal, the "R/R" draws attention to it. This succinctly indicates to the performer that something unusual is happening.



Full-reversal: pro to pro

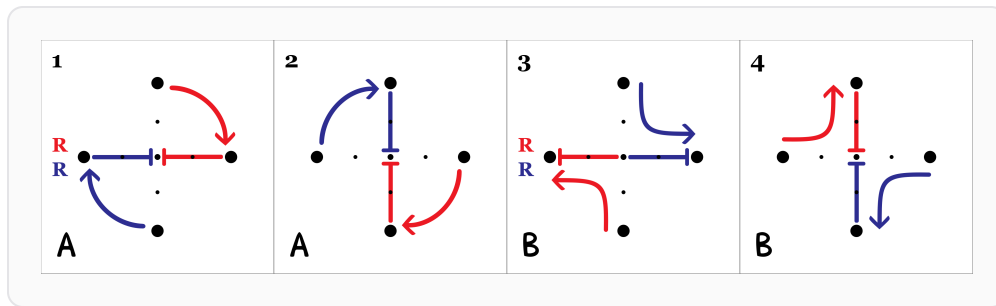


Full-reversal: anti to anti

Examples — AABB

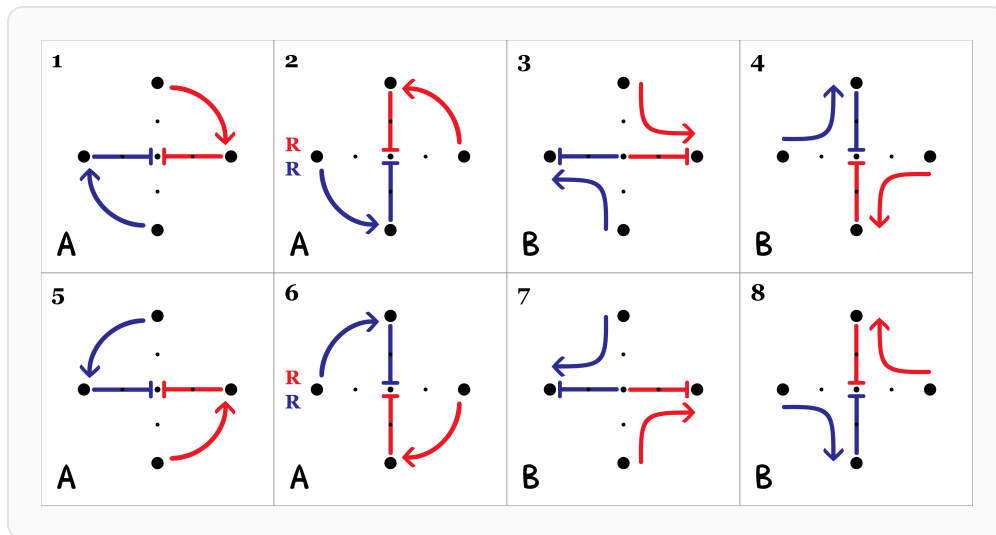
Let's practice reversals and permutations. We'll use AABB as an example to explore different reversal placements. These start from the same alpha start position. Interpret it from the first motions.

Here's an AABB in which both staves execute prop-reversals after beats 2 and 4, notated by an "R/R" in between the pictographs. This requires negative space or a body turn to execute.



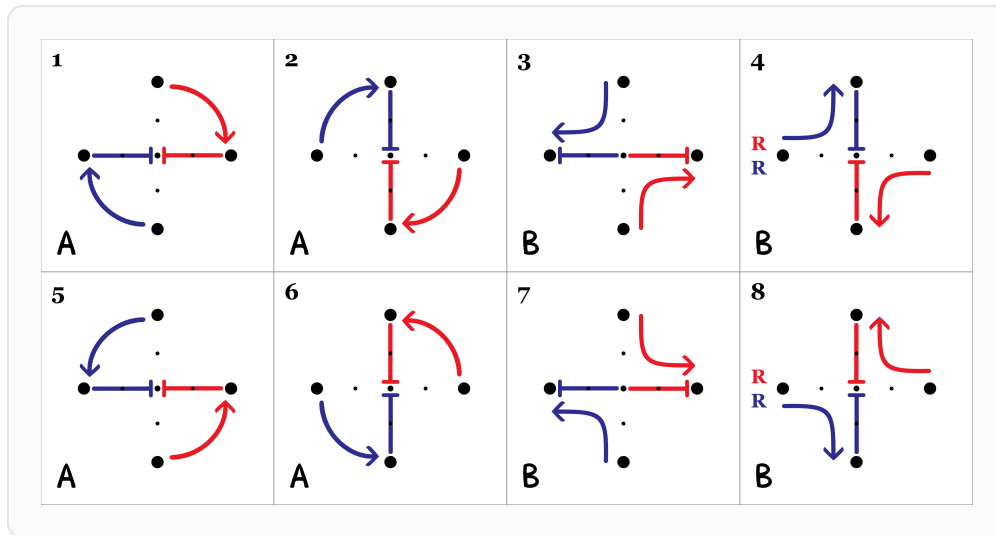
AABB with prop-reversals after beats 2 and 4

Let's place the reversals in a different place. This time we'll put them after beat 1. This will put our left hand on top after beat 4, so we'll repeat the sequence again mirrored (with a reversal after beat 5) to return to our original home position. This is a Mirrored LOOP.



AABB Mirrored LOOP

Now let's look at another variation of AABB×2 with reversals after beats 3 & 7.

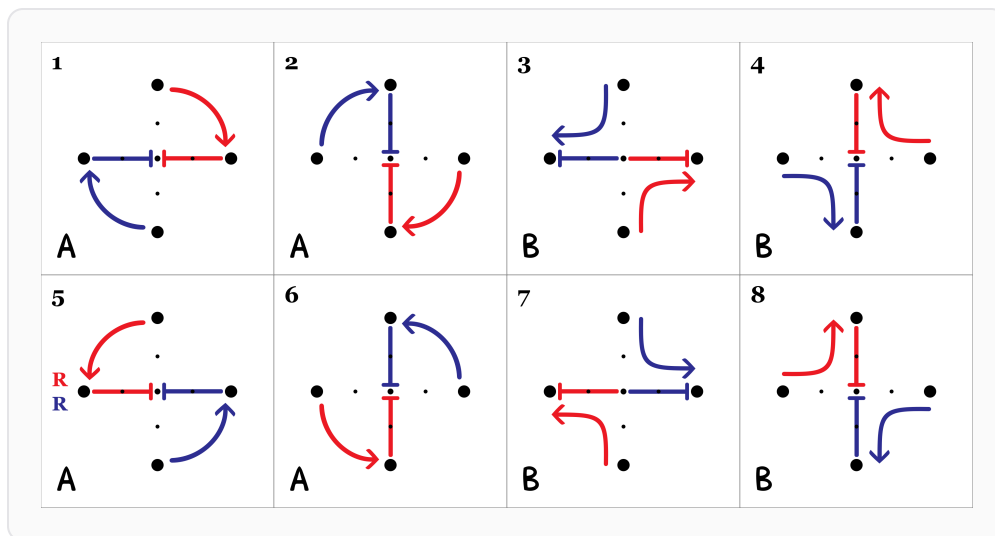


AABB×2 with reversals after beats 3 & 7

As demonstrated with these examples, a reversal in different locations in the word can lead to a notably different outcome. The word AABBB is not limited to one presentation, it is a broad category of sequences that includes those letters with variations on reversals and thumb orientation.

AABB Body Turn / CCCC

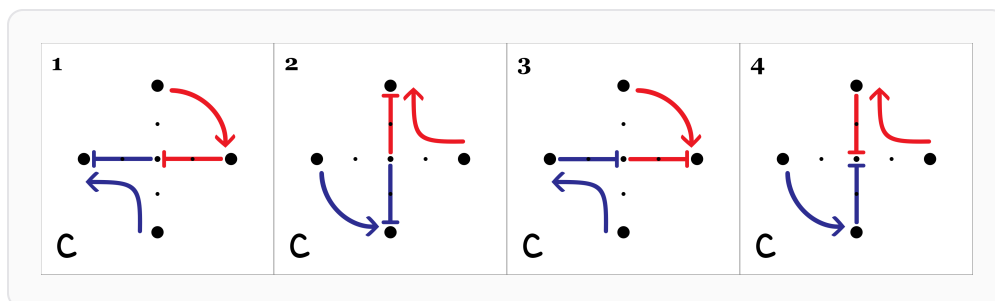
Let's add reversals after the second B. Take note — beat 5 has the right hand coming down on the left side of the grid, so it's impossible to follow through to beat 6 while remaining square with the audience in wall plane. We must body turn on beat 5. If turning left, we can bring the left staff into the plane behind us as it comes up, moving our relative position into wheel plane. This sequence is a good example of how body turns can serve both as a method of motion execution and a body movement that can add energy and contrast.



AABB with body turn

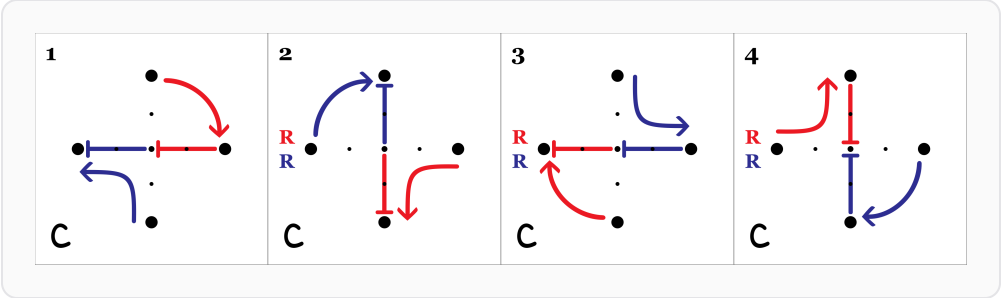
Now let's observe how reversals affect pro/anti hybrid words like CCCC. They present more variations than non-hybrids:

Hand-reversal



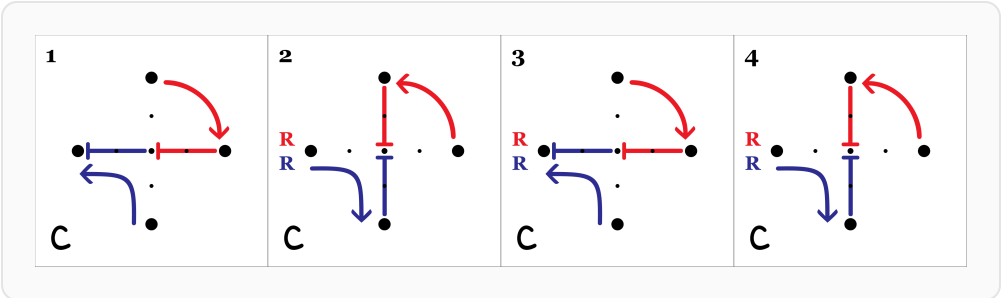
CCCC — Hand-reversal

Prop-reversal



CCCC — Prop-reversal

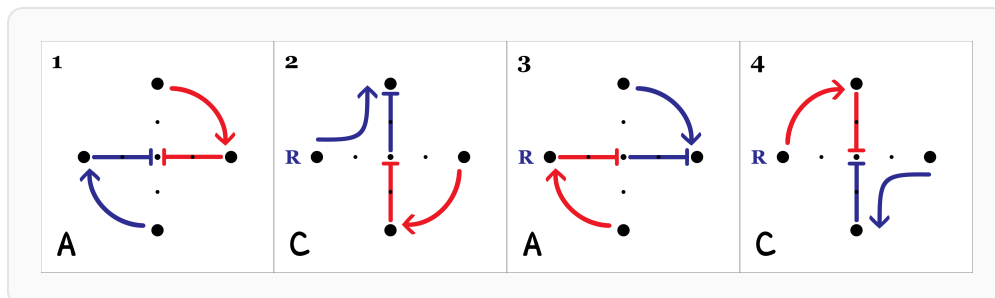
Full-reversal



CCCC — Full-reversal

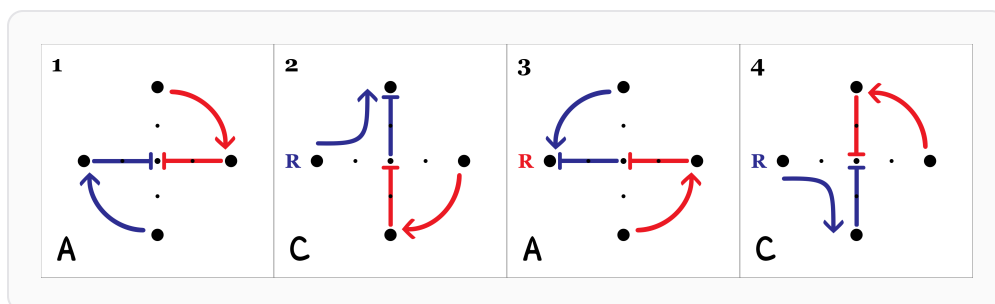
ACAC / BCBC

When combining a hybrid like C with a non-hybrid like A or B, a prop-reversal is necessary. Let's look at the word ACAC. In this variation, the left hand does a reversal on every beat. Give it a try.



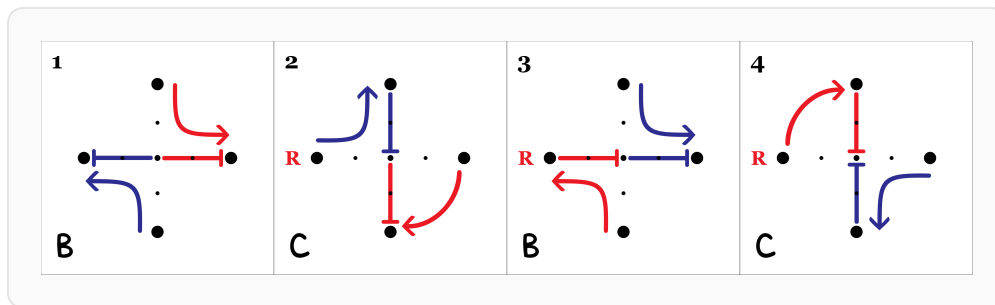
ACAC with prop-reversals

It would be impossible to execute ACAC without using a prop-reversal. The previous example shows the hands moving in a continuous path. Let's change that in the next example by including a full-reversal in the middle. In this example, the reversals alternate between left (R) and right (R). This example uses every type of reversal — hand, prop, and full. Challenge yourself to identify where each one occurs.



ACAC with hand, prop, and full reversals

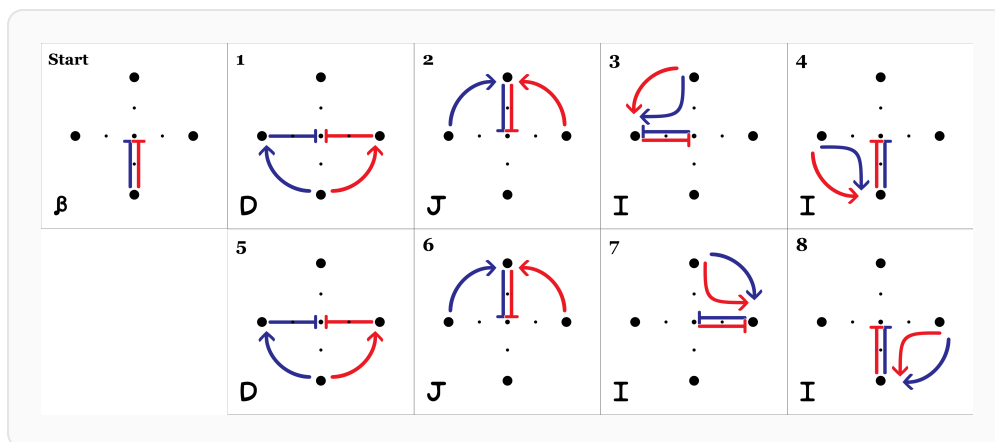
Prop-reversals are also required with BCBC, as shown in this example. Here, the right hand is prop-reversing after every beat. Eventually, it returns to home.



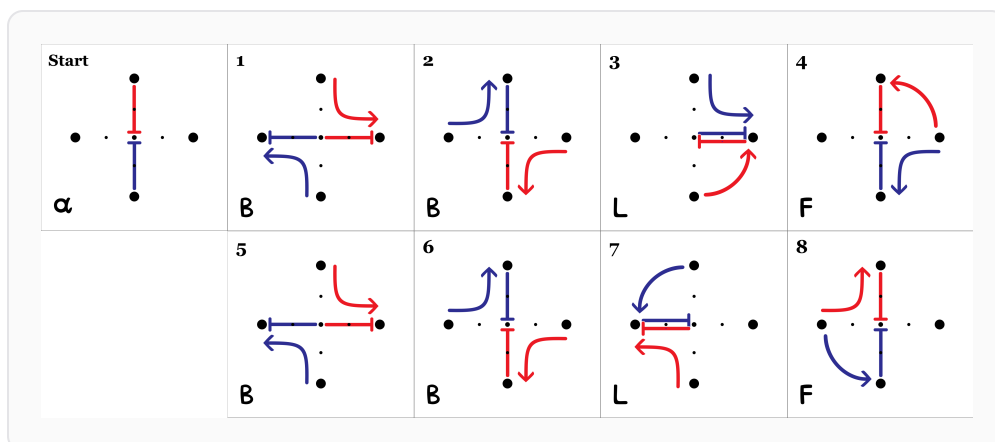
BCBC with prop-reversals

Type 1 LOOPS

In this example of DJII, the graphs in the second repetition (beats 5-8) mirror the graphs in the first repetition (beats 1-4), classifying it as a Mirrored LOOP.

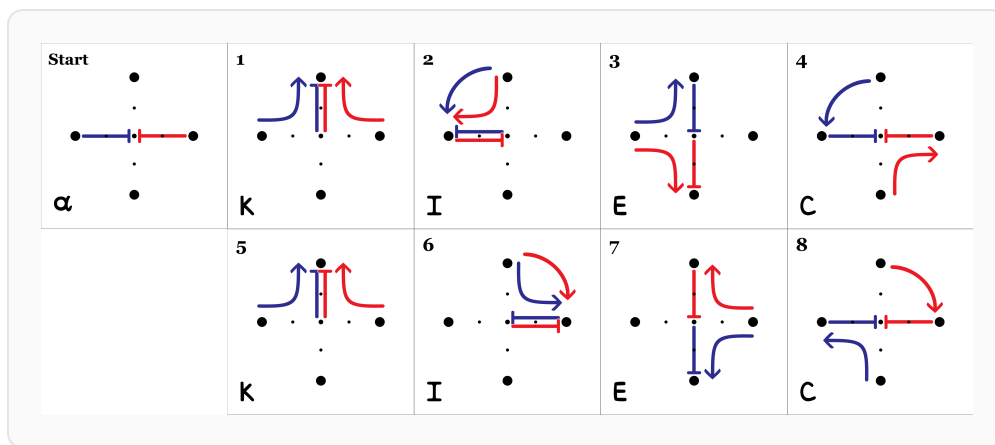


DJII — Mirrored LOOP



BBLF — Swapped & Rotated LOOP

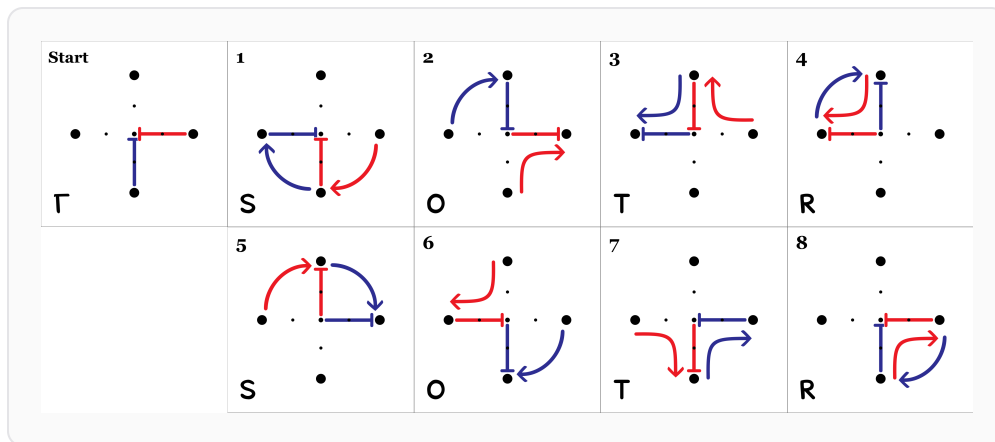
In this example of KIEC, the colors are swapped in the second half, so it is classified as a Swapped & Mirrored LOOP.



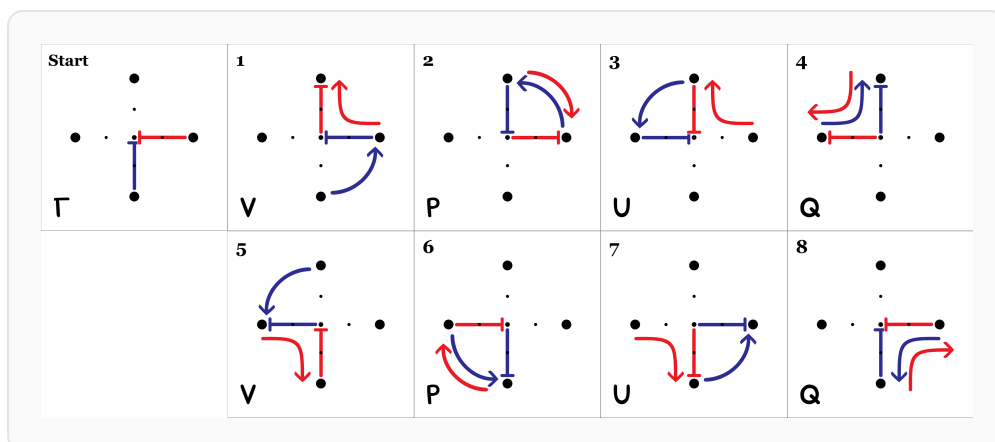
KIEC — Swapped & Mirrored LOOP

Gamma LOOPS

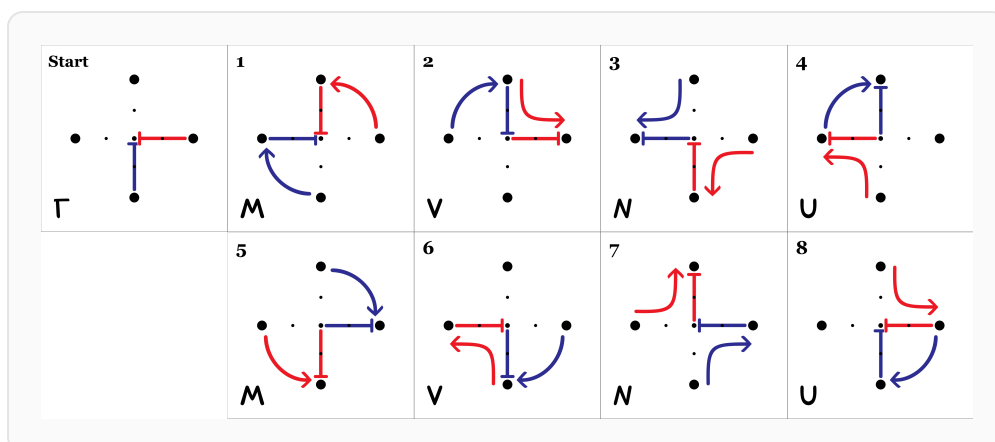
The Γ letters can connect to any other Γ letter. In these examples, each word ends in gamma position on the opposite side. By repeating the word from there, we return to home position.



SOTR — Rotated LOOP



VPUQ — Rotated LOOP

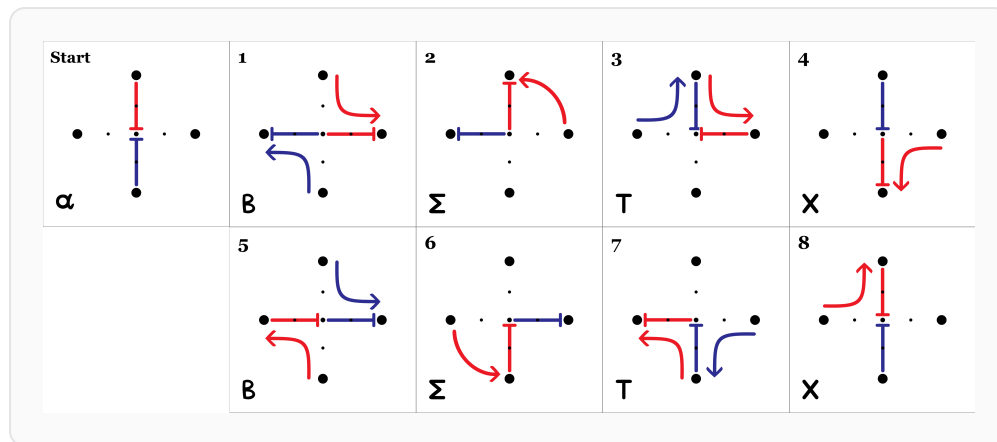


MVNU — Rotated LOOP

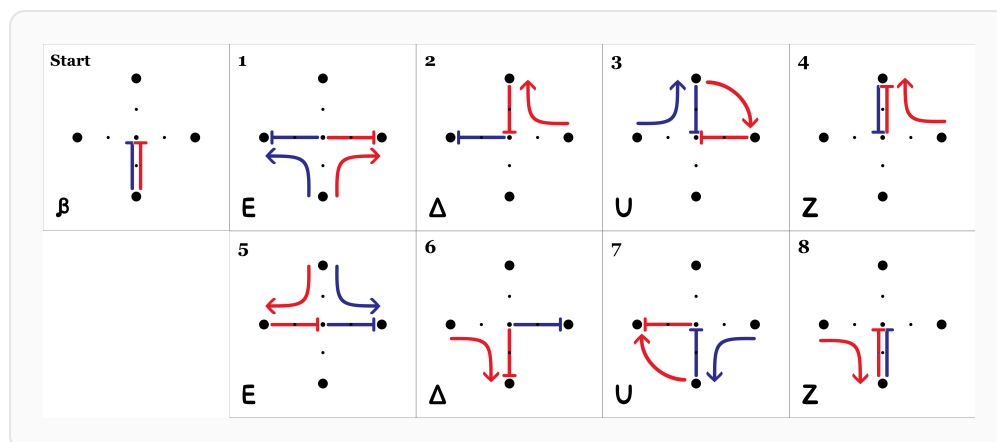
Note that the pictographs in each second word repetition are rotated 180°. Because of this, these examples are classified as Rotated LOOPs.

Type 2 LOOPS

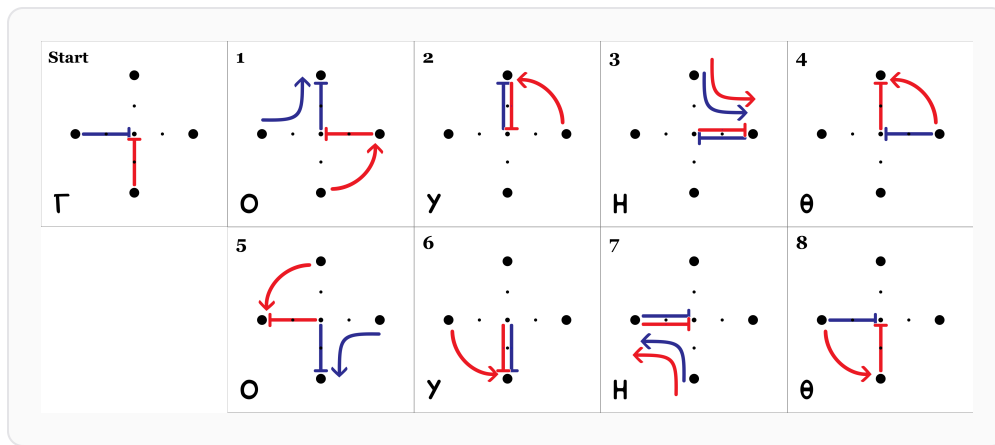
These words use the Type 2 letters to travel between α/β and Γ . Since each repetition is rotated by 180°, these are all Rotated LOOPS.



$\beta\Sigma TX$ — Rotated LOOP



$E\Delta UZ$ — Rotated LOOP

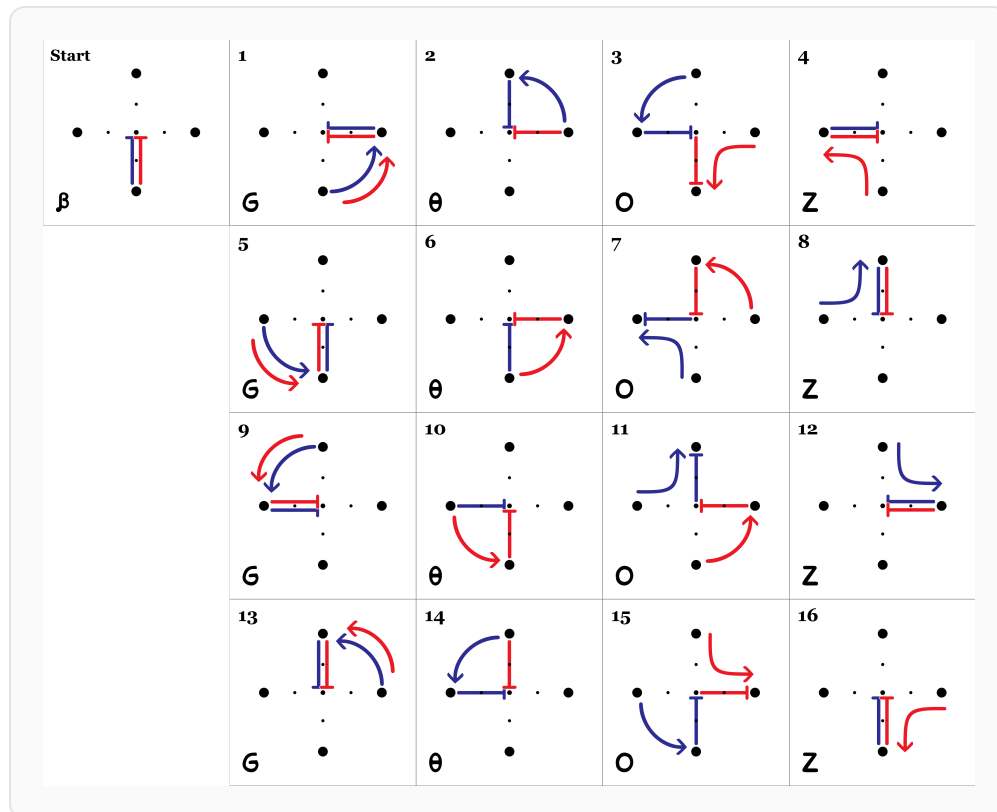


OYHθ — Rotated LOOP

16-Count Sequences

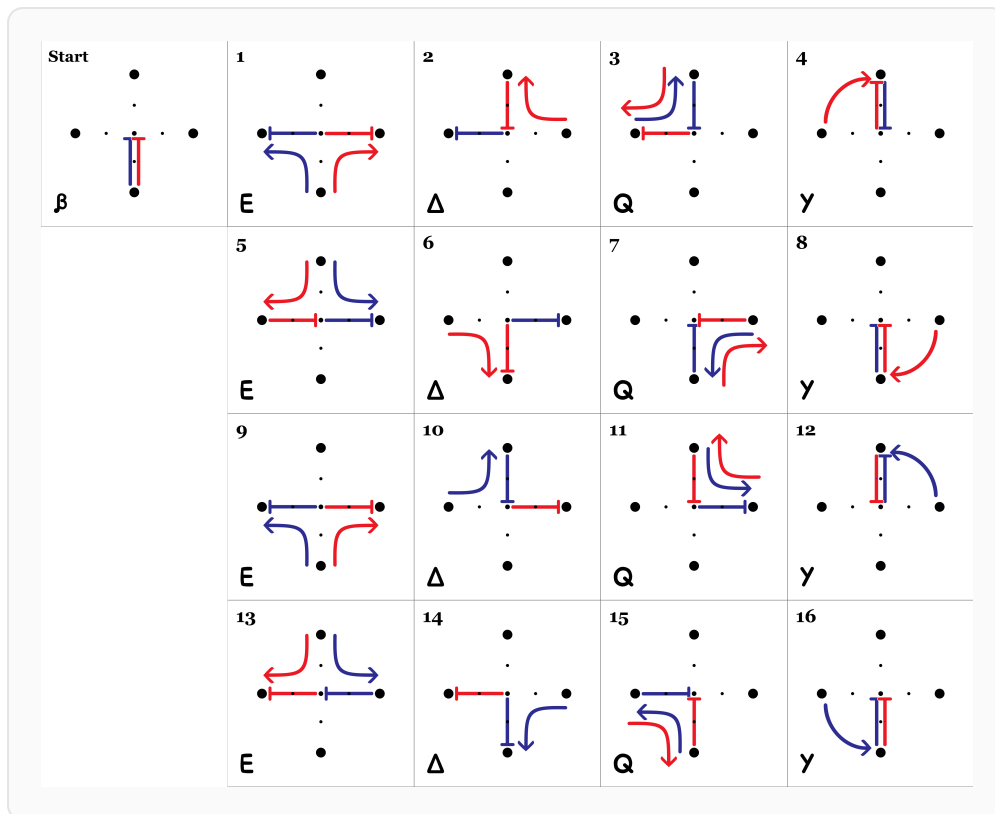
These 4-letter words repeat 4 times, giving us 16-count sequences.

GθOZ — Here, each repetition of the word ends in a Γ that is 90° from its start. This means it will take 4 repetitions to return to home. (Rotated+Swapped LOOP)



GθOZ — Rotated+Swapped LOOP

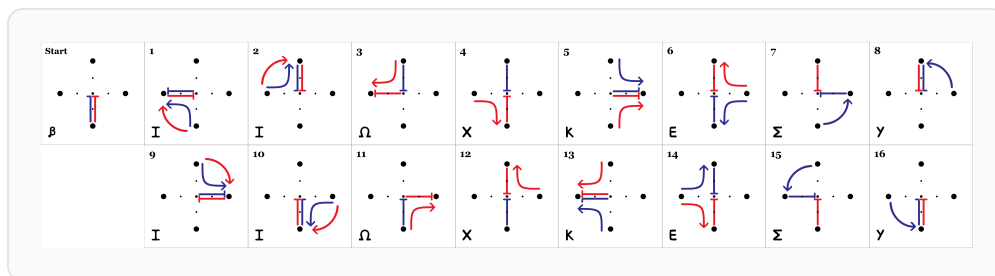
EΔQY — Here, the staves return to home after two word repetitions. To make it symmetrical, it repeats twice more, filling the rest of the quadrants. (Rotated + Mirrored + Swapped LOOP)



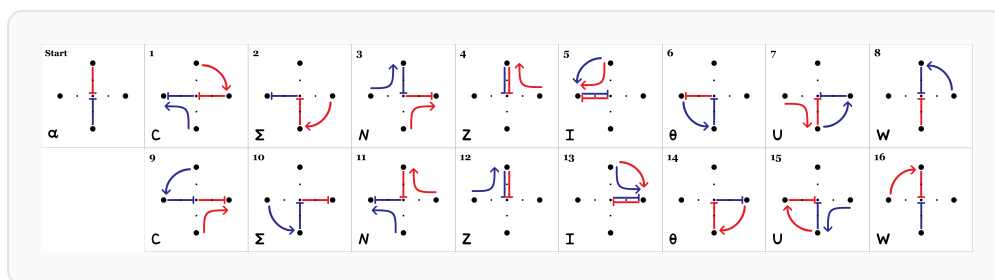
EΔQY — Rotated + Mirrored + Swapped LOOP

8-Letter Words

Words can be any length. These 8-letter words repeat twice, to create 16-count sequences.



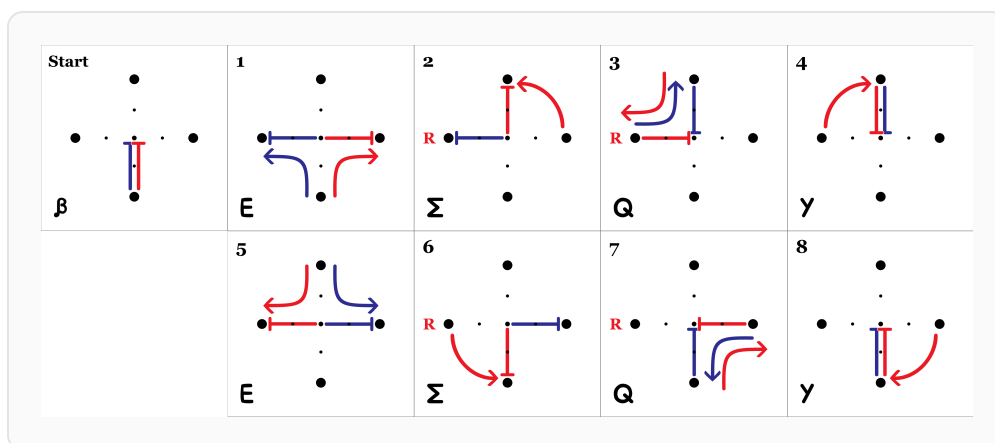
IIΩXKEΣY — Rotated LOOP



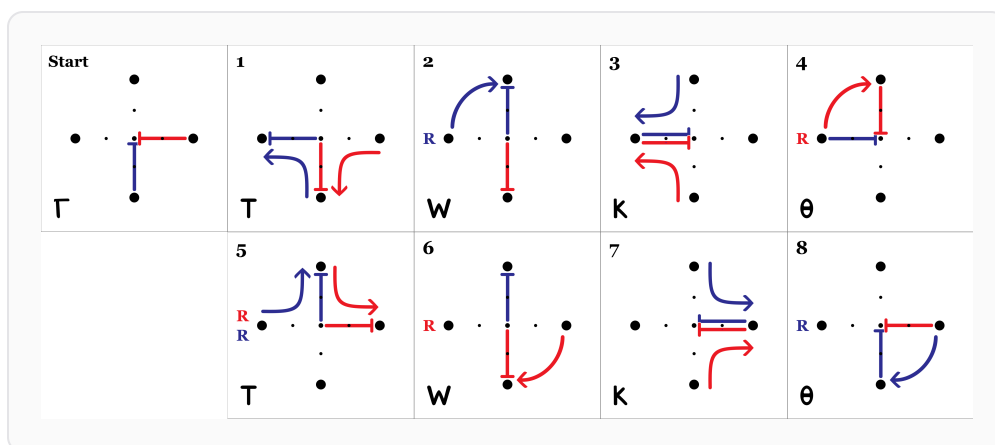
CSNZIθUW — Mirrored + Swapped LOOP

Prop-Reversal LOOPS

Each of these words uses a prop-reversal. These examples are Rotated LOOPS.

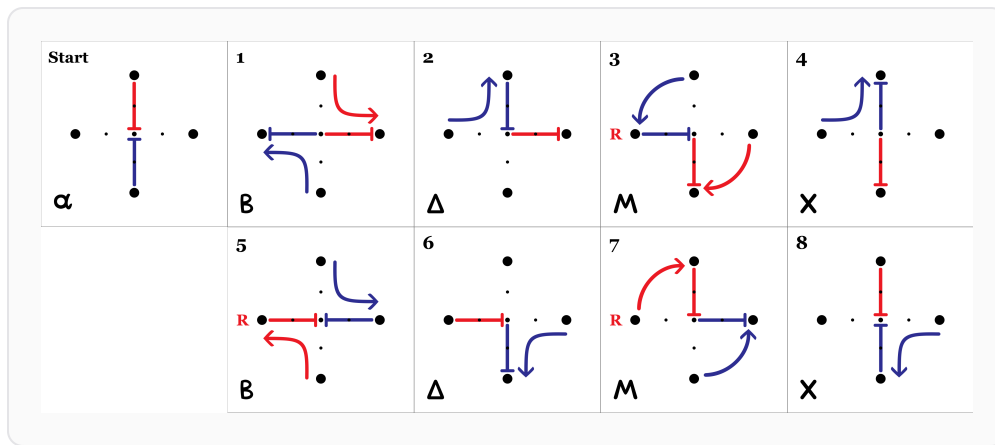


EΣQY — Rotated LOOP



TWKθ — Rotated LOOP

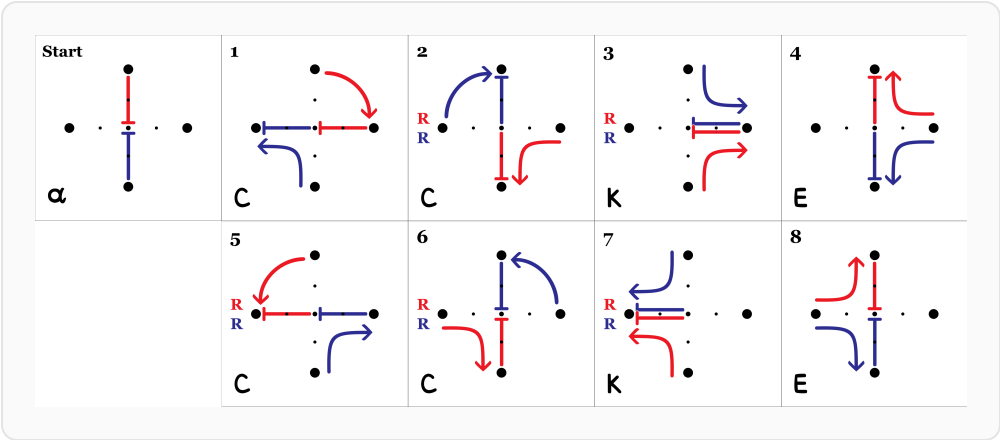
In this example of BΔMX, the right hand stops on beats 2 and 4 before resuming its motion around the center point. Even when there is a beat with no motion in between, we can still mark the reversal with an "R" to indicate the prop reversal.



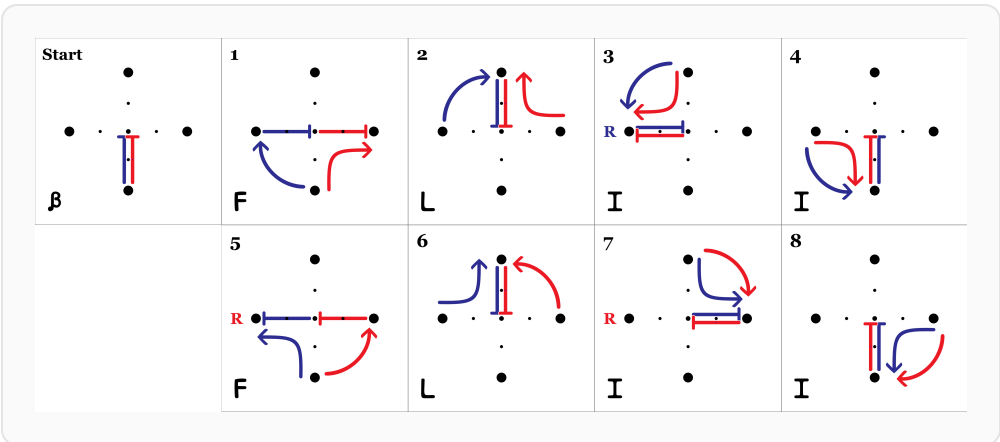
B Δ MX — Rotated LOOP

Full-Reversal LOOPS

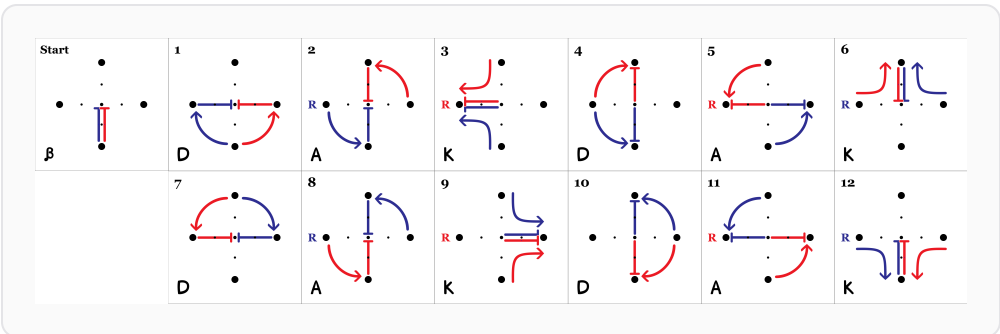
Each of these words uses a full-reversal. There are also prop-reversals within these words. Challenge yourself to identify each one.



CCKE



FLII



DAK